Notes on the Classical Theory of Linear Models

1 The Multiple Linear Regression Model

1.1 Model Specification and Matrix Notation

Definition 1.1 (Multiple Linear Regression Model)**.** Given n observations, each consisting of a dependent variable Y_i and k independent variables $X_{i1}, \ldots, X_{ik}$, the **multiple linear regression model** assumes

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \ldots, n, \tag{1}$$

where β_0 is the intercept, $\beta_1, \ldots, \beta_k$ are slope coefficients, and ε_i is the error term. Equivalently, in matrix form,

$$\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}} \tag{2}$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}_{n\times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{21} & \cdots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{nk} \end{pmatrix}_{n\times(k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{(k+1)\times 1}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}_{n\times 1}.$$

The matrix $\mathbf{X}$ is called the **design matrix**. Its first column is the all-ones vector $\mathbf{1} = (1, 1, \ldots, 1)'$, corresponding to the intercept. We write $\mathbf{x}_i'$ for the i-th row of $\mathbf{X}$, so that $Y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$.

1.2 Classical Assumptions

Definition 1.2 (Classical Assumptions)**.** The following assumptions are imposed on model (2):

(A1) **Linearity**: $\mathsf{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$.

(A2) **Zero conditional mean**: $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$.

(A3) **Spherical errors**: $\mathrm{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$. This comprises two sub-conditions:

- *Homoscedasticity*: $\mathrm{var}(\varepsilon_i) = \sigma^2$ for all i (a constant).
- *No serial correlation*: $\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.

(A4) **Full rank**: $\mathrm{rank}(\mathbf{X}) = k + 1$ (the columns of $\mathbf{X}$ are linearly independent). This requires $n \geqslant k + 1$. For variance estimation and inference (Sections 3–7), we additionally require $n > k + 1$ so that $\mathrm{MSE} = \mathrm{SSE}/(n - k - 1)$ (see Definition 3.1) is well-defined.

(A5) **Normality**: $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Assumptions (A1)–(A4) are called the **Gauss–Markov assumptions**. Assumption (A5) is additionally needed for exact distributional results in finite samples.

Remark. Throughout this article, we treat $\mathbf{X}$ as fixed (non-stochastic) and condition on it. All expectations, variances, and distributions are conditional on $\mathbf{X}$, but we suppress the conditioning notation for brevity.

Remark (Relationship between (A1) and (A2))**.** Given the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, assumptions (A1) and (A2) are logically equivalent: taking conditional expectations of both sides gives $\mathsf{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta} + \mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}]$, so $\mathsf{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$ holds if and only if $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$. We state both because each form is convenient in different proofs: (A1) is closer to how the model is written, and (A2) is closer to how residuals are handled. Conditioning on $\mathbf{X}$ (previous remark) also means that inside any proof, $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ can be used without further notation—these are the conditional expectation and variance with the conditioning suppressed.

2 Ordinary Least Squares

2.1 Definition and the Normal Equations

Definition 2.1 (Ordinary Least Squares)**.** The **OLS estimator** $\hat{\boldsymbol{\beta}}$ is the value of $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$Q(\boldsymbol{\beta}) = \sum_{i=1}^{n} (Y_i - \mathbf{x}_i'\boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2. \tag{3}$$

Lemma 2.1 (Matrix Differentiation Formulas)**.** For $\boldsymbol{\beta} \in \mathbb{R}^p$, constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p\times p}$, write $Df(\boldsymbol{\beta}) := \partial f/\partial \boldsymbol{\beta}'$ (numerator layout, row vector). Then

(i) $D(\mathbf{a}'\boldsymbol{\beta}) = D(\boldsymbol{\beta}'\mathbf{a}) = \mathbf{a}'$;

(ii) $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$, which equals $2\boldsymbol{\beta}'\mathbf{A}$ when $\mathbf{A}' = \mathbf{A}$.

Proof. Apply the *first identification theorem*: $\mathrm{d}f(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'\mathrm{d}\boldsymbol{\beta} \iff Df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'$.

(i). $\mathrm{d}(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'\mathrm{d}\boldsymbol{\beta}$ (since $\mathrm{d}\mathbf{a} = 0$), so $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$; the case $D(\boldsymbol{\beta}'\mathbf{a})$ follows from $\boldsymbol{\beta}'\mathbf{a} = \mathbf{a}'\boldsymbol{\beta}$.

(ii). Since $\mathrm{d}\mathbf{A} = 0$, the product rule and the scalar identity $(\mathrm{d}\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'\mathrm{d}\boldsymbol{\beta}$ give

$$\mathrm{d}(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = (\mathrm{d}\boldsymbol{\beta})'\mathbf{A}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}\,\mathrm{d}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{A}'\mathrm{d}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{A}\,\mathrm{d}\boldsymbol{\beta} = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')\,\mathrm{d}\boldsymbol{\beta}.$$

□

2.2 Derivation via Differentiation

Theorem 2.1 (OLS Estimator)**.** Under assumption (A4), the OLS estimator is

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.} \tag{4}$$

Proof. Step 1: Expand $Q(\boldsymbol{\beta})$.

$$\begin{aligned} Q(\boldsymbol{\beta}) &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{Y}'\mathbf{Y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned} \tag{5}$$

In the last line, $\mathbf{Y}'\mathbf{X}\boldsymbol{\beta}$ is a scalar, hence equals its transpose $\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y}$.

Step 2: First-order condition. Apply Lemma 2.1 term by term, with $\mathbf{a} = \mathbf{X}'\mathbf{Y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric):

$$DQ(\boldsymbol{\beta}) = -2(\mathbf{X}'\mathbf{Y})' + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}.$$

Equivalently, by the differentials in the proof of Lemma 2.1, $\mathrm{d}Q = (-2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X})\,\mathrm{d}\boldsymbol{\beta}$, and the first identification theorem reads off DQ. Setting $DQ = \mathbf{0}'$ and transposing gives the **normal equation**

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}. \tag{6}$$

Step 3: Solve. By (A4), $\mathrm{rank}(\mathbf{X}) = k + 1$. We show $\mathrm{rank}(\mathbf{X}'\mathbf{X}) = \mathrm{rank}(\mathbf{X})$: $\mathbf{X}\mathbf{v} = \mathbf{0}$ trivially implies $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$; conversely, $\mathbf{X}'\mathbf{X}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 = 0$, hence $\mathbf{X}\mathbf{v} = \mathbf{0}$. So $\mathbf{X}$ and $\mathbf{X}'\mathbf{X}$ have the same null space, hence the same rank. Thus $\mathbf{X}'\mathbf{X}$ is $(k+1)\times(k+1)$ of full rank, hence invertible. Multiplying (6) by $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Step 4: Verify the second-order condition. The Hessian matrix is

$$\frac{\partial^2 Q}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} = 2\mathbf{X}'\mathbf{X}.$$

Under assumption (A4), for any nonzero $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0$. Hence $\mathbf{X}'\mathbf{X}$ is positive definite, confirming that $\hat{\boldsymbol{\beta}}$ is a global minimum. □

2.3 Derivation via Orthogonal Projection

Theorem 2.2 (OLS as Orthogonal Projection)**.** The fitted vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$, and the OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

Proof. The OLS problem is to minimize $Q(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$, equivalently, to find the point in $\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{k+1}\}$ closest to $\mathbf{Y}$ in Euclidean norm.

Step 1: Orthogonality characterization of the minimum. We claim $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimizes $\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2$ over $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ iff $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

($\Leftarrow$) Assume the orthogonality condition and let $\tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{X})$ is a subspace, $\hat{\mathbf{Y}} - \tilde{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$, so $(\mathbf{Y} - \hat{\mathbf{Y}}) \perp (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$. Pythagoras applied to $\mathbf{Y} - \tilde{\mathbf{Y}} = (\mathbf{Y} - \hat{\mathbf{Y}}) + (\hat{\mathbf{Y}} - \tilde{\mathbf{Y}})$ gives

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 \;\geqslant\; \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2,$$

with equality iff $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}}$. So $\hat{\mathbf{Y}}$ is the unique minimizer.

($\Rightarrow$) Let $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ minimize. Suppose, for contradiction, some $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ has $\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$; normalize to $\|\mathbf{v}\|^2 = 1$. For $\tilde{\mathbf{Y}} = \hat{\mathbf{Y}} + t\mathbf{v} \in \mathcal{C}(\mathbf{X})$,

$$\|\mathbf{Y} - \tilde{\mathbf{Y}}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - 2t\,\mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) + t^2,$$

minimized at $t^* = \mathbf{v}'(\mathbf{Y} - \hat{\mathbf{Y}}) \neq 0$ with value $\|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 - (t^*)^2 < \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2$, contradicting optimality. Hence $\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X})$.

Step 2: Translate the orthogonality condition to a normal equation. The residual is orthogonal to every vector in $\mathcal{C}(\mathbf{X})$ if and only if it is orthogonal to every column of $\mathbf{X}$ (since those columns span $\mathcal{C}(\mathbf{X})$):

$$\mathbf{Y} - \hat{\mathbf{Y}} \perp \mathcal{C}(\mathbf{X}) \iff \mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}.$$

Writing $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ for some $\hat{\boldsymbol{\beta}}$, this becomes $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, the same normal equation (6) obtained by differentiation.

Step 3: Solve. Under assumption (A4), $\mathbf{X}'\mathbf{X}$ is invertible, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, and the minimizer $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ exists and is unique. □

2.4 Hat Matrix, Residual Maker, and Idempotent Matrices

Definition 2.2 (Hat Matrix and Residual Maker Matrix)**.** The **hat matrix** is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \tag{7}$$

The **residual maker matrix** (or **annihilator matrix**) is

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \tag{8}$$

Then $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\mathbf{e} = \mathbf{M}\mathbf{Y}$.

Definition 2.3 (Idempotent Matrix)**.** A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.

Lemma 2.2 (Trace Equals Rank for Idempotent Matrices)**.** If $\mathbf{A}$ is an $n \times n$ idempotent matrix, then $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$.

Proof. We show every idempotent matrix is diagonalizable with eigenvalues in $\{0, 1\}$.

Step 1: Eigenvalues are in $\{0, 1\}$. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\mathbf{v} = \mathbf{A}\mathbf{v} = \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$, forcing $\lambda \in \{0, 1\}$.

Step 2: $\mathbb{R}^n = \mathcal{N}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})$. Any $\mathbf{v} \in \mathbb{R}^n$ decomposes as $\mathbf{v} = (\mathbf{v} - \mathbf{A}\mathbf{v}) + \mathbf{A}\mathbf{v}$, with $\mathbf{A}\mathbf{v} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} - \mathbf{A}\mathbf{v} \in \mathcal{N}(\mathbf{A})$ (since $\mathbf{A}(\mathbf{v} - \mathbf{A}\mathbf{v}) = \mathbf{A}\mathbf{v} - \mathbf{A}^2\mathbf{v} = \mathbf{0}$). The sum is direct: if $\mathbf{v} = \mathbf{A}\mathbf{w} \in \mathcal{C}(\mathbf{A}) \cap \mathcal{N}(\mathbf{A})$, then $\mathbf{v} = \mathbf{A}\mathbf{w} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{v} = \mathbf{0}$.

Step 3: Diagonalizability. Any $\mathbf{u} \in \mathcal{C}(\mathbf{A})$ satisfies $\mathbf{A}\mathbf{u} = \mathbf{u}$ ($\mathbf{u} = \mathbf{A}\mathbf{w}$, so $\mathbf{A}\mathbf{u} = \mathbf{A}^2\mathbf{w} = \mathbf{A}\mathbf{w} = \mathbf{u}$). Combining a basis of $\mathcal{N}(\mathbf{A})$ (eigenvalue 0) and a basis of $\mathcal{C}(\mathbf{A})$ (eigenvalue 1) gives an eigenvector basis of $\mathbb{R}^n$ by Step 2. Thus $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $\mathbf{\Lambda}$ diagonal and entries in $\{0, 1\}$.

Step 4: Conclude. Let $r = \operatorname{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Then $\mathbf{\Lambda}$ has r ones and $n - r$ zeros, so by the cyclic property of trace,

$$\operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}) = \operatorname{tr}(\mathbf{\Lambda}) = r = \operatorname{rank}(\mathbf{A}). \qquad \square$$

Theorem 2.3 (Properties of $\mathbf{H}$ and $\mathbf{M}$)**.** The matrices $\mathbf{H}$ and $\mathbf{M}$ satisfy:

(i) $\mathbf{H}$ and $\mathbf{M}$ are both symmetric and idempotent.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $\mathbf{M}\mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}$.

(iv) $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Proof. (i) Symmetry of H: Since $\mathbf{X}'\mathbf{X}$ is symmetric, so is $(\mathbf{X}'\mathbf{X})^{-1}$:

$$\mathbf{H}' = [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Idempotency of H:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\underbrace{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{\mathbf{I}_{k+1}}\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}.$$

Symmetry and idempotency of M: $\mathbf{M}' = (\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}' = \mathbf{I} - \mathbf{H} = \mathbf{M}$. Also, $\mathbf{M}^2 = (\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H} = \mathbf{I} - \mathbf{H} = \mathbf{M}$.

(ii) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X}\mathbf{I}_{k+1} = \mathbf{X}$, and $\mathbf{M}\mathbf{X} = (\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(iii) $\mathbf{H}\mathbf{M} = \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{H} - \mathbf{H}^2 = \mathbf{H} - \mathbf{H} = \mathbf{0}$, and $\mathbf{M}\mathbf{H} = (\mathbf{I} - \mathbf{H})\mathbf{H} = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$.

(iv) By Lemma 2.2, $\operatorname{tr}(\mathbf{H}) = \operatorname{rank}(\mathbf{H})$. We show $\operatorname{rank}(\mathbf{H}) = k + 1$ in two steps.

Step 1: $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. The inclusion $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ is immediate: every column of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is of the form $\mathbf{X}\mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^{k+1}$. Conversely, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ because $\mathbf{H}\mathbf{X} = \mathbf{X}$ by (ii), so every column of $\mathbf{X}$ equals $\mathbf{H}$ times a column of $\mathbf{X}$, hence lies in $\mathcal{C}(\mathbf{H})$.

Step 2: $\dim \mathcal{C}(\mathbf{X}) = k + 1$ by assumption (A4).

Combining, $\operatorname{rank}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{H}) = \dim \mathcal{C}(\mathbf{X}) = k + 1$, so $\operatorname{tr}(\mathbf{H}) = k + 1$. Finally, $\operatorname{tr}(\mathbf{M}) = \operatorname{tr}(\mathbf{I}) - \operatorname{tr}(\mathbf{H}) = n - (k + 1) = n - k - 1$. $\square$

Remark ($\mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$)**.** Combining (i), (ii), and the identity $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ proved above: $\mathbf{H}$ is symmetric, idempotent, and has column space $\mathcal{C}(\mathbf{X})$. These three properties characterize the *orthogonal projection* onto $\mathcal{C}(\mathbf{X})$. Geometrically, $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ is the closest point in $\mathcal{C}(\mathbf{X})$ to $\mathbf{Y}$ in Euclidean distance (this is the content of §2, specifically the orthogonal projection derivation). Analogously, $\mathbf{M} = \mathbf{I} - \mathbf{H}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})^{\perp}$.

2.5 Errors Versus Residuals

Remark (Errors versus residuals)**.** The **error** $\varepsilon_i = Y_i - \mathbf{x}_i'\boldsymbol{\beta}$ is the deviation from the true (unknown) regression function; it is *unobservable*. The **residual** $e_i = Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}} = Y_i - \hat{Y}_i$ is the deviation from the fitted regression function; it is *observable*. In matrix form, $\mathbf{e} = \mathbf{M}\mathbf{Y} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}$ (since $\mathbf{M}\mathbf{X} = \mathbf{0}$ by Theorem 2.3(ii)). The residual vector is a linear transformation of the error vector.

2.6 Algebraic Properties of OLS Residuals

Theorem 2.4 (Implications of the Normal Equations)**.** The OLS residuals satisfy:

(i) $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (residuals are orthogonal to every column of $\mathbf{X}$).

(ii) $\sum_{i=1}^n e_i = 0$ (residuals sum to zero), provided the model includes an intercept.

(iii) $\hat{\mathbf{Y}}'\mathbf{e} = 0$ (residuals are orthogonal to fitted values).

Proof. (i) From the normal equation $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, rearranging:

$$\mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \iff \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \iff \mathbf{X}'\mathbf{e} = \mathbf{0}.$$

(ii) The first column of $\mathbf{X}$ is $\mathbf{1}$. The first component of the vector equation $\mathbf{X}'\mathbf{e} = \mathbf{0}$ is $\mathbf{1}'\mathbf{e} = \sum_{i=1}^n e_i = 0$. This relies on the presence of the intercept column.

(iii) Since $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\hat{\mathbf{Y}}'\mathbf{e} = (\mathbf{X}\hat{\boldsymbol{\beta}})'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = \hat{\boldsymbol{\beta}}' \cdot \mathbf{0} = 0.$$

Geometrically, $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{e} \perp \mathcal{C}(\mathbf{X})$. □

2.7 Leverages

Definition 2.4 (Leverage)**.** The **leverage** of the i-th observation is h_{ii}, the i-th diagonal entry of $\mathbf{H}$.

Theorem 2.5 (Properties of Leverages)**.**

(i) $0 \leqslant h_{ii} \leqslant 1$ for all i.

(ii) $\sum_{i=1}^n h_{ii} = k + 1$.

Proof. (i) Since $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H} = \mathbf{H}^2 = \mathbf{H}'\mathbf{H}$. Comparing the (i, i) entry on both sides using $h_{ji} = h_{ij}$:

$$h_{ii} = \sum_{j=1}^n h_{ij}h_{ji} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2.$$

The rightmost sum is $\geqslant 0$, so $h_{ii} \geqslant h_{ii}^2$, which rearranges to $h_{ii}(1 - h_{ii}) \geqslant 0$. The leftmost expression $h_{ii} = \sum_j h_{ij}^2 \geqslant 0$ is also a sum of squares. Combining $h_{ii} \geqslant 0$ with $h_{ii}(1 - h_{ii}) \geqslant 0$ forces $h_{ii} \leqslant 1$.

(ii) $\sum_{i=1}^n h_{ii} = \operatorname{tr}(\mathbf{H}) = k + 1$ by Theorem 2.3(iv). □

3 Finite-Sample Properties Without Normality

In this section, only assumptions (A1)–(A4) are used. No distributional assumption on $\boldsymbol{\varepsilon}$ is needed.

Before deriving the finite-sample properties of $\hat{\boldsymbol{\beta}}$, we collect the notation for the sum-of-squares and mean-square quantities used throughout the remaining sections.

Definition 3.1 (Sums of Squares and Mean Squares)**.** Given the fitted model $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, define:

- **Total sum of squares**: $\mathrm{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2$, where $\bar{Y} = n^{-1}\sum_i Y_i$.

- **Regression sum of squares**: $\mathrm{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$.
- **Error (residual) sum of squares**: $\mathrm{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$, where $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$.
- **Mean squared error**: $\mathrm{MSE} = s^2 = \mathrm{SSE}/(n-k-1)$.
- **Mean squared regression**: $\mathrm{MSR} = \mathrm{SSR}/k$.

The decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ is proved in Theorem 6.1 (§6); the unbiasedness $\mathsf{E}[\mathrm{MSE}] = \sigma^2$ is proved in Theorem 3.4 below.

We also record a tool used repeatedly in this and later sections.

Lemma 3.1 (Variance under a linear transformation)**.** Let $\mathbf{z}$ be an $n \times 1$ random vector with $\mathrm{var}(\mathbf{z})$ defined, $\mathbf{A}$ an $m \times n$ constant matrix, and $\mathbf{b} \in \mathbb{R}^m$ a constant vector. Then

$$\mathrm{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) = \mathbf{A}\ \mathrm{var}(\mathbf{z})\,\mathbf{A}'. \tag{9}$$

In particular, under assumptions (A1)–(A3), $\mathrm{var}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$.

Proof. Let $\boldsymbol{\mu} = \mathsf{E}[\mathbf{z}]$. Since $\mathbf{A}$ and $\mathbf{b}$ are constant, $\mathsf{E}[\mathbf{A}\mathbf{z} + \mathbf{b}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$. Therefore

$$\begin{aligned}
\mathrm{var}(\mathbf{A}\mathbf{z} + \mathbf{b}) &= \mathsf{E}\big[(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})(\mathbf{A}\mathbf{z} + \mathbf{b} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})'\big] \\
&= \mathsf{E}\big[\mathbf{A}(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})'\mathbf{A}'\big] \\
&= \mathbf{A}\,\mathsf{E}[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})']\,\mathbf{A}' = \mathbf{A}\ \mathrm{var}(\mathbf{z})\,\mathbf{A}',
\end{aligned}$$

since constant matrices pull outside the expectation. Applying this to $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ (with $\mathbf{A} = \mathbf{I}_n$, $\mathbf{b} = \mathbf{X}\boldsymbol{\beta}$ constant given the conditioning on $\mathbf{X}$) gives $\mathrm{var}(\mathbf{Y}) = \mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ by (A3). □

3.1 Unbiasedness

Theorem 3.1 (Unbiasedness of $\hat{\boldsymbol{\beta}}$)**.** Under (A1)–(A2), the OLS estimator is unbiased:

$$\boxed{\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}.} \tag{10}$$

Proof. Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$:

$$\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \\
&= \mathbf{I}_{k+1}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \\
&= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}.
\end{aligned} \tag{11}$$

Taking expectations, using (A2) ($\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathsf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0} = \boldsymbol{\beta}.$$ □

3.2 Variance–Covariance Matrix

Theorem 3.2 (Variance–Covariance Matrix of $\hat{\boldsymbol{\beta}}$)**.** Under (A1)–(A4):

$$\boxed{\mathrm{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.} \tag{12}$$

Proof. From (11), $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, which is a linear transformation of $\boldsymbol{\varepsilon}$ (with a constant shift). Applying Lemma 3.1 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$:

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\ \mathrm{var}(\boldsymbol{\varepsilon})\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

By (A3), $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$. Substituting:

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1},$$

using $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. The diagonal entry $[\mathrm{var}(\hat{\boldsymbol{\beta}})]_{jj} = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ gives $\mathrm{var}(\hat{\beta}_j)$. □

3.3 The Gauss–Markov Theorem

Theorem 3.3 (Gauss–Markov Theorem)**.** Under (A1)–(A4), the OLS estimator $\hat{\boldsymbol{\beta}}$ is the **Best Linear Unbiased Estimator** (BLUE) of $\boldsymbol{\beta}$. That is, for any other linear unbiased estimator $\tilde{\boldsymbol{\beta}} = \mathbf{CY}$ (where $\mathbf{C}$ is a $(k+1) \times n$ constant matrix), the difference $\mathrm{var}(\tilde{\boldsymbol{\beta}}) - \mathrm{var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite. In particular, $\mathrm{var}(\tilde{\beta}_j) \geqslant \mathrm{var}(\hat{\beta}_j)$ for every j.

Proof. Step 1: Necessary condition for unbiasedness. By (A1), $\mathsf{E}[\tilde{\boldsymbol{\beta}}] = \mathbf{CX}\boldsymbol{\beta} = \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$ forces $\mathbf{CX} = \mathbf{I}_{k+1}$ (setting $\mathbf{v} = \mathbf{e}_j$ in $\mathbf{CXv} = \mathbf{v}$ reads off the j-th column).

Step 2: Decompose C. Write $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$, where $\mathbf{D}$ is the excess beyond the OLS coefficient matrix. From $\mathbf{CX} = \mathbf{I}_{k+1}$:

$$\mathbf{DX} = \mathbf{CX} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_{k+1} - \mathbf{I}_{k+1} = \mathbf{0}.$$

Step 3: Compute $\mathrm{var}(\tilde{\boldsymbol{\beta}})$**.** Since $\tilde{\boldsymbol{\beta}} = \mathbf{CY}$ with $\mathbf{C}$ constant, Lemma 3.1 gives $\mathrm{var}(\tilde{\boldsymbol{\beta}}) = \mathbf{C}\,\mathrm{var}(\mathbf{Y})\mathbf{C}' = \sigma^2\mathbf{CC}'$:

$$\mathrm{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}\big]\big[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'\big].$$

Expanding yields four terms:

$$\begin{aligned}
\sigma^2\mathbf{CC}' = {} & \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} && \text{(term A)}\\
& + \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}' && \text{(term B)}\\
& + \sigma^2\mathbf{DX}(\mathbf{X}'\mathbf{X})^{-1} && \text{(term C)}\\
& + \sigma^2\mathbf{DD}'. && \text{(term D)}
\end{aligned}$$

Term A simplifies to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ because $\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_{k+1}$. Term C vanishes because $\mathbf{DX} = \mathbf{0}$ (Step 2). Term B equals (term C)$'$, so it also vanishes. Hence:

$$\mathrm{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{DD}' = \mathrm{var}(\hat{\boldsymbol{\beta}}) + \sigma^2\mathbf{DD}'.$$

Since $\mathbf{DD}'$ is positive semi-definite (for any $\mathbf{v} \in \mathbb{R}^{k+1}$, $\mathbf{v}'\mathbf{DD}'\mathbf{v} = \|\mathbf{D}'\mathbf{v}\|^2 \geqslant 0$):

$$\mathrm{var}(\tilde{\boldsymbol{\beta}}) - \mathrm{var}(\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{DD}' \succeq \mathbf{0}.$$

Equality holds if and only if $\mathbf{D}'\mathbf{v} = \mathbf{0}$ for all $\mathbf{v}$, i.e., $\mathbf{D} = \mathbf{0}$; equivalently, $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$. □

3.4 Estimation of σ^2

Lemma 3.2 (Expected Value of a Quadratic Form)**.** Let $\mathbf{z}$ be an $n \times 1$ random vector with $\mathsf{E}[\mathbf{z}] = \boldsymbol{\mu}$ and $\mathrm{var}(\mathbf{z}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be an $n \times n$ constant matrix. Then:

$$\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}. \tag{13}$$

In particular, if $\mathsf{E}[\mathbf{z}] = \mathbf{0}$, then $\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})$.

Proof. The quadratic form is a scalar: $\mathbf{z}'\mathbf{Az} = \sum_{i,j} a_{ij}z_iz_j$. Taking expectations:

$$\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \sum_{i,j} a_{ij}\mathsf{E}[z_iz_j].$$

From $\mathrm{cov}(z_i, z_j) = \mathsf{E}[z_iz_j] - \mu_i\mu_j$, we get $\mathsf{E}[z_iz_j] = \Sigma_{ij} + \mu_i\mu_j$. Substituting:

$$\mathsf{E}[\mathbf{z}'\mathbf{Az}] = \sum_{i,j} a_{ij}\Sigma_{ij} + \sum_{i,j} a_{ij}\mu_i\mu_j.$$

The first sum equals $\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma})$: indeed, $\mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}) = \sum_i(\mathbf{A}\boldsymbol{\Sigma})_{ii} = \sum_i\sum_j a_{ij}\Sigma_{ji} = \sum_{i,j} a_{ij}\Sigma_{ij}$, where the last step uses the symmetry $\Sigma_{ji} = \Sigma_{ij}$. The second sum equals $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$: indeed, $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu} = \sum_i \mu_i(\mathbf{A}\boldsymbol{\mu})_i = \sum_i \mu_i \sum_j a_{ij}\mu_j = \sum_{i,j} a_{ij}\mu_i\mu_j$. Combining yields (13). □

Theorem 3.4 (Unbiasedness of MSE)**.** Under (A1)–(A4), the **mean squared error**

$$s^2 = \text{MSE} = \frac{\text{SSE}}{n-k-1} = \frac{\mathbf{e}'\mathbf{e}}{n-k-1} \tag{14}$$

is an unbiased estimator of σ^2: $\mathsf{E}[\text{MSE}] = \sigma^2$.

Proof. Step 1: Express SSE **as a quadratic form in** $\boldsymbol{\varepsilon}$**.** Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ (Remark after Definition 2.2):

$$\text{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}'\mathbf{M}\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon},$$

where the last step uses $\mathbf{M}'\mathbf{M} = \mathbf{M}\mathbf{M} = \mathbf{M}$ (symmetric idempotent).

Step 2: Apply Lemma 3.2. With $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{A} = \mathbf{M}$, $\boldsymbol{\mu} = \mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ (by (A2)), and $\boldsymbol{\Sigma} = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ (by (A3)):

$$\mathsf{E}[\text{SSE}] = \mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}] = \text{tr}(\mathbf{M}\cdot\sigma^2\mathbf{I}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma^2\,\text{tr}(\mathbf{M}).$$

Step 3: Evaluate the trace. By Theorem 2.3(iv), $\text{tr}(\mathbf{M}) = n-k-1$. Hence $\mathsf{E}[\text{SSE}] = \sigma^2(n-k-1)$.

Step 4: Conclude.

$$\mathsf{E}[\text{MSE}] = \mathsf{E}\left[\frac{\text{SSE}}{n-k-1}\right] = \frac{\sigma^2(n-k-1)}{n-k-1} = \sigma^2. \qquad \square$$

3.5 Variance of Individual Residuals

Theorem 3.5 (Variance of Residuals)**.** Under (A1)–(A4):

$$\text{var}(\mathbf{e}) = \sigma^2\mathbf{M}, \qquad \text{and in particular, } \text{var}(e_i) = \sigma^2(1-h_{ii}). \tag{15}$$

Proof. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ and $\mathbf{M}$ is symmetric ($\mathbf{M}' = \mathbf{M}$) and idempotent ($\mathbf{M}^2 = \mathbf{M}$):

$$\text{var}(\mathbf{e}) = \mathbf{M}\,\text{var}(\boldsymbol{\varepsilon})\mathbf{M}' = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}' = \sigma^2\mathbf{M}\mathbf{M} = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}.$$

The (i,i) entry gives $\text{var}(e_i) = \sigma^2 m_{ii}$. Since $\mathbf{M} = \mathbf{I} - \mathbf{H}$, we have $m_{ii} = 1 - h_{ii}$, so $\text{var}(e_i) = \sigma^2(1-h_{ii})$. $\square$

Remark. By Theorem 2.5(i), $0 \leqslant h_{ii} \leqslant 1$, so $\text{var}(e_i) \leqslant \sigma^2$. High-leverage observations (large h_{ii}) produce residuals with smaller variance: the fitted line is pulled toward them, leaving less room for the residual to vary.

4 Multivariate Normal Tools

This section collects the distributional results needed for exact inference in Section 5.

Theorem 4.1 (Closure Under Linear Transformations)**.** If $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is an $n \times 1$ normal random vector, $\mathbf{A}$ is an $m \times n$ constant matrix, and $\mathbf{b} \in \mathbb{R}^m$ is a constant vector, then

$$\mathbf{A}\mathbf{z} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'). \tag{16}$$

Proof. The moment-generating function (MGF) of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \exp(\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t})$. The MGF of $\mathbf{w} = \mathbf{A}\mathbf{z} + \mathbf{b}$ is:

$$\begin{aligned} M_{\mathbf{w}}(\mathbf{t}) &= \mathsf{E}\left[e^{\mathbf{t}'(\mathbf{A}\mathbf{z}+\mathbf{b})}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot \mathsf{E}\left[e^{(\mathbf{A}'\mathbf{t})'\mathbf{z}}\right] = e^{\mathbf{t}'\mathbf{b}} \cdot M_{\mathbf{z}}(\mathbf{A}'\mathbf{t}) \\ &= e^{\mathbf{t}'\mathbf{b}} \cdot \exp\big((\mathbf{A}'\mathbf{t})'\boldsymbol{\mu} + \tfrac{1}{2}(\mathbf{A}'\mathbf{t})'\boldsymbol{\Sigma}(\mathbf{A}'\mathbf{t})\big) \\ &= \exp\big(\mathbf{t}'(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \tfrac{1}{2}\mathbf{t}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'\mathbf{t}\big). \end{aligned}$$

This is the MGF of $N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$. By uniqueness of the MGF, the result follows. $\square$

Theorem 4.2 (Uncorrelatedness $\Leftrightarrow$ Independence Under Joint Normality)**.** Let $(\mathbf{U}', \mathbf{V}')'$ be jointly normal:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_U \\ \boldsymbol{\mu}_V \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{UU} & \boldsymbol{\Sigma}_{UV} \\ \boldsymbol{\Sigma}_{VU} & \boldsymbol{\Sigma}_{VV} \end{pmatrix} \right).$$

Then $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$ if and only if $\mathbf{U}$ and $\mathbf{V}$ are independent.

Proof. ($\Leftarrow$): Independence implies uncorrelatedness for any distribution.

($\Rightarrow$): With $\boldsymbol{\Sigma}_{UV} = \mathbf{0}$, the covariance is block-diagonal. The joint MGF is $M_{(\mathbf{U},\mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp\big((\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} + \frac{1}{2}(\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma}(\mathbf{s}', \mathbf{t}')'\big)$. Block expansion gives

$$(\mathbf{s}', \mathbf{t}')\boldsymbol{\mu} = \mathbf{s}'\boldsymbol{\mu}_U + \mathbf{t}'\boldsymbol{\mu}_V, \qquad (\mathbf{s}', \mathbf{t}')\boldsymbol{\Sigma} \begin{pmatrix} \mathbf{s} \\ \mathbf{t} \end{pmatrix} = \mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s} + \mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t},$$

using $\boldsymbol{\Sigma}_{UV} = \boldsymbol{\Sigma}_{VU}' = \mathbf{0}$. The joint MGF factors:

$$M_{(\mathbf{U},\mathbf{V})}(\mathbf{s}, \mathbf{t}) = \exp\big(\mathbf{s}'\boldsymbol{\mu}_U + \tfrac{1}{2}\mathbf{s}'\boldsymbol{\Sigma}_{UU}\mathbf{s}\big) \cdot \exp\big(\mathbf{t}'\boldsymbol{\mu}_V + \tfrac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}_{VV}\mathbf{t}\big) = M_{\mathbf{U}}(\mathbf{s})M_{\mathbf{V}}(\mathbf{t}),$$

which implies independence. □

Remark. For general (non-normal) distributions, uncorrelatedness does *not* imply independence. The equivalence is a special property of the multivariate normal.

Theorem 4.3 (Chi-Square Distribution of Quadratic Forms)**.** Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{A}$ be an $n \times n$ symmetric idempotent matrix. Then

$$\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi^2_r, \quad \text{where } r = \operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}). \tag{17}$$

Proof. Step 1: Spectral decomposition. Since $\mathbf{A}$ is symmetric, there exists an orthogonal matrix $\mathbf{P}$ ($\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}$) such that $\mathbf{A} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'$, where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. By idempotency and Lemma 2.2, each $\lambda_i \in \{0, 1\}$. Without loss of generality, let the first r eigenvalues be 1 and the rest be 0:

$$\boldsymbol{\Lambda} = \operatorname{diag}(\underbrace{1, \ldots, 1}_{r}, \underbrace{0, \ldots, 0}_{n-r}).$$

Step 2: Apply orthogonal transformation. Let $\mathbf{w} = \mathbf{P}'\mathbf{z}$. Since $\mathbf{P}$ is orthogonal:

$$\mathsf{E}[\mathbf{w}] = \mathbf{P}'\mathsf{E}[\mathbf{z}] = \mathbf{0}, \qquad \operatorname{var}(\mathbf{w}) = \mathbf{P}'\mathbf{I}\mathbf{P} = \mathbf{I}_n.$$

By Theorem 4.1, $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_n)$, so $w_1, \ldots, w_n$ are i.i.d. $N(0, 1)$.

Step 3: Evaluate the quadratic form.

$$\mathbf{z}'\mathbf{A}\mathbf{z} = \mathbf{z}'\mathbf{P}\boldsymbol{\Lambda}\mathbf{P}'\mathbf{z} = (\mathbf{P}'\mathbf{z})'\boldsymbol{\Lambda}(\mathbf{P}'\mathbf{z}) = \mathbf{w}'\boldsymbol{\Lambda}\mathbf{w} = \sum_{i=1}^{n} \lambda_i w_i^2 = \sum_{i=1}^{r} w_i^2.$$

Step 4: Conclude. Since $w_1, \ldots, w_r$ are independent $N(0, 1)$, by definition $\sum_{i=1}^{r} w_i^2 \sim \chi^2_r$. □

Theorem 4.4 (Cochran's Theorem — Independence of Quadratic Forms)**.** Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}_1, \mathbf{A}_2$ be $n \times n$ symmetric idempotent matrices with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Then $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent.

Proof. Define $\mathbf{U} = \mathbf{A}_1\mathbf{z}$ and $\mathbf{V} = \mathbf{A}_2\mathbf{z}$. Stack them into a single vector obtained from $\mathbf{z}$ by one linear map:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{V} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \mathbf{z}.$$

Since $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$, Theorem 4.1 implies $(\mathbf{U}', \mathbf{V}')'$ is *jointly* normal. Their cross-covariance is:

$$\operatorname{cov}(\mathbf{U}, \mathbf{V}) = \mathbf{A}_1 \operatorname{var}(\mathbf{z}) \mathbf{A}_2' = \mathbf{A}_1 \mathbf{I} \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A}_2 = \mathbf{0},$$

where $\mathbf{A}_2' = \mathbf{A}_2$ by symmetry. By Theorem 4.2, $\mathbf{U}$ and $\mathbf{V}$ are independent. By symmetric idempotency of $\mathbf{A}_1$, $\mathbf{z}'\mathbf{A}_1\mathbf{z} = \mathbf{z}'\mathbf{A}_1'\mathbf{A}_1\mathbf{z} = \|\mathbf{U}\|^2$ is a function of $\mathbf{U}$ alone; similarly $\mathbf{z}'\mathbf{A}_2\mathbf{z} = \|\mathbf{V}\|^2$ depends only on $\mathbf{V}$. The two quadratic forms are therefore independent. □

Remark. In Section 5, Cochran's theorem is applied with $\mathbf{A}_1 = \mathbf{H} - \mathbf{H}_0$ and $\mathbf{A}_2 = \mathbf{M}$ to prove SSR $\perp$ SSE.

5 Exact Inference Under Normality

From this section onward, all five assumptions (A1)–(A5) are in force.

5.1 Sampling Distribution of $\hat{\boldsymbol{\beta}}$

Theorem 5.1 (Normality of $\hat{\boldsymbol{\beta}}$)**.** Under (A1)–(A5):

$$\boxed{\hat{\boldsymbol{\beta}} \sim N\big(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big)\,.} \tag{18}$$

Proof. From (11), $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$. This is an affine transformation of $\boldsymbol{\varepsilon}$. Under (A5), $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. By Theorem 4.1 with $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, $\mathbf{z} = \boldsymbol{\varepsilon}$, $\mathbf{b} = \boldsymbol{\beta}$:

$$\begin{aligned}\hat{\boldsymbol{\beta}} &\sim N\big(\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \mathbf{0},\ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\big)\\ &= N\big(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big).\end{aligned}$$

□

5.2 Distribution of SSE/σ^2 and Independence from $\hat{\boldsymbol{\beta}}$

Theorem 5.2 (Chi-Square Distribution of SSE/σ^2)**.** Under (A1)–(A5):

$$\frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-k-1}, \tag{19}$$

and SSE is independent of $\hat{\boldsymbol{\beta}}$.

Proof. Step 1: Standardize. Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Since $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$:

$$\frac{\text{SSE}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'\mathbf{M}\mathbf{z}.$$

Step 2: Apply Theorem 4.3. The matrix $\mathbf{M}$ is symmetric and idempotent (Theorem 2.3(i)), with $\operatorname{tr}(\mathbf{M}) = n - k - 1$ (Theorem 2.3(iv)). Hence $\mathbf{z}'\mathbf{M}\mathbf{z} \sim \chi^2_{n-k-1}$.

Step 3: Prove independence. Stack $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ into a single vector obtained from $\boldsymbol{\varepsilon}$ by one linear map:

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \mathbf{M} \end{pmatrix} \boldsymbol{\varepsilon}.$$

Since $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, Theorem 4.1 makes this stacked vector *jointly* normal (individual normality alone would not suffice; joint normality requires that both pieces come from a single normal vector). Their cross-covariance is:

$$\begin{aligned}\operatorname{cov}\big((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon},\ \mathbf{M}\boldsymbol{\varepsilon}\big) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \operatorname{var}(\boldsymbol{\varepsilon})\mathbf{M}'\\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{M} = \mathbf{0},\end{aligned}$$

using $\mathbf{X}'\mathbf{M} = (\mathbf{M}\mathbf{X})' = \mathbf{0}$ (Theorem 2.3(ii)). By Theorem 4.2, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\mathbf{e}$ are independent. Since $\hat{\boldsymbol{\beta}}$ is $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ plus a constant and SSE $= \mathbf{e}'\mathbf{e}$ is a function of $\mathbf{e}$, $\hat{\boldsymbol{\beta}}$ is independent of $\mathbf{e}$ (hence of SSE). □

5.3 OLS as Maximum Likelihood Under Normality

Theorem 5.3 (OLS Equals MLE Under Normality)**.** Under (A1)–(A5), the OLS estimator $\hat{\boldsymbol{\beta}}$ is also the maximum likelihood estimator of $\boldsymbol{\beta}$. The MLE of σ^2 is $\hat{\sigma}^2_{\text{MLE}} = \text{SSE}/n$, which is biased.

Proof. Under (A5), $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, so by Theorem 4.1 applied to the affine transformation $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. The log-likelihood is:

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

Maximizing over $\boldsymbol{\beta}$: In $\ell(\boldsymbol{\beta}, \sigma^2)$, the first two terms do not involve $\boldsymbol{\beta}$, and the last term is $-(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})/(2\sigma^2)$ with $\sigma^2 > 0$ treated as fixed. Since $-1/(2\sigma^2) < 0$, maximizing ℓ over $\boldsymbol{\beta}$ is equivalent to minimizing $(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = Q(\boldsymbol{\beta})$. This is exactly the OLS problem, so $\hat{\boldsymbol{\beta}}_{\text{MLE}} = \hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

Maximizing over σ^2: Substituting $\hat{\boldsymbol{\beta}}$ and noting that $(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \text{SSE}$, the profile log-likelihood becomes

$$\ell(\hat{\boldsymbol{\beta}}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{\text{SSE}}{2\sigma^2}.$$

Differentiating with respect to σ^2:

$$\frac{\partial\ell}{\partial\sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2(\sigma^2)^2} = 0 \implies \hat{\sigma}^2_{\text{MLE}} = \frac{\text{SSE}}{n}.$$

Second-order condition: Writing $u = \sigma^2$ for brevity,

$$\frac{\partial^2\ell}{\partial u^2} = \frac{n}{2u^2} - \frac{\text{SSE}}{u^3}.$$

At the critical point $\hat{u} = \text{SSE}/n$, so $u^2 = \text{SSE}^2/n^2$ and $u^3 = \text{SSE}^3/n^3$:

$$\left.\frac{\partial^2\ell}{\partial u^2}\right|_{\hat{u}} = \frac{n \cdot n^2}{2\text{SSE}^2} - \frac{\text{SSE} \cdot n^3}{\text{SSE}^3} = \frac{n^3}{2\text{SSE}^2} - \frac{n^3}{\text{SSE}^2} = -\frac{n^3}{2\text{SSE}^2} < 0,$$

confirming a maximum.

Bias of $\hat{\sigma}^2_{\text{MLE}}$. Using Theorem 3.4 ($\mathsf{E}[\text{SSE}] = (n-k-1)\sigma^2$):

$$\mathsf{E}[\hat{\sigma}^2_{\text{MLE}}] = \mathsf{E}\left[\frac{\text{SSE}}{n}\right] = \frac{(n-k-1)\sigma^2}{n} = \frac{n-k-1}{n}\sigma^2 \;<\; \sigma^2,$$

so $\hat{\sigma}^2_{\text{MLE}}$ underestimates σ^2 on average by a factor of $(n-k-1)/n$. The unbiased estimator is $\text{MSE} = \text{SSE}/(n-k-1)$, which rescales by this factor. □

6 Sum of Squares and the Coefficient of Determination

6.1 Variance Decomposition

Definition 6.1 (Centering Matrix)**.** The **centering matrix** is $\mathbf{M}_0 = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$. For any vector $\mathbf{v}$, $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$, where $\bar{v} = \frac{1}{n}\mathbf{1}'\mathbf{v}$.

Remark (Notation: $\mathbf{M}_0$ versus $\mathbf{M}$)**.** The subscript "0" in $\mathbf{M}_0$ signals that it is the residual maker with respect to the *intercept only* (the single column $\mathbf{1}$), whereas $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ from Definition 2.2 is the residual maker with respect to the *full* design matrix $\mathbf{X}$. Conceptually,

$$\mathbf{M}_0 \text{ centers a vector,} \qquad \mathbf{M} \text{ regresses out all of } \mathbf{X}.$$

Both are symmetric idempotent, but they project onto different subspaces:

$$\mathbf{M}_0 : \mathbb{R}^n \to \mathbf{1}^\perp, \qquad \mathbf{M} : \mathbb{R}^n \to \mathcal{C}(\mathbf{X})^\perp.$$

If the model includes an intercept then $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, hence $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathbf{1}^\perp$, and so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}$ (this follows from the projection identity $\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0\mathbf{H} = \mathbf{H}_0$ proved in Lemma 7.1, Step 2, by taking complements). In particular, $\mathbf{M}$ and $\mathbf{M}_0$ are not equal, nor is one a simple scalar multiple of the other; they coincide only in the trivial case $\mathbf{X} = \mathbf{1}$.

Remark. $\mathbf{M}_0$ is symmetric and idempotent (being the residual maker for the intercept-only model), with $\operatorname{tr}(\mathbf{M}_0) = n - 1$. Also define $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}' = \mathbf{I} - \mathbf{M}_0$, the projection onto $\mathcal{C}(\mathbf{1})$.

Theorem 6.1 (Variance Decomposition)**.** In the model with an intercept, the sums of squares introduced in Definition 3.1 satisfy:

$$\boxed{\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE},} \tag{20}$$

with matrix expressions:

- $\mathrm{SST} = \sum(Y_i - \bar{Y})^2 = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$ (Total Sum of Squares),
- $\mathrm{SSR} = \sum(\hat{Y}_i - \bar{Y})^2 = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$ (Regression Sum of Squares),
- $\mathrm{SSE} = \sum(Y_i - \hat{Y}_i)^2 = \mathbf{e}'\mathbf{e}$ (Error/Residual Sum of Squares).

Proof. Step 1: Verify the matrix expressions for SST **and** SSR**.** For any vector $\mathbf{v} \in \mathbb{R}^n$, Definition 6.1 gives $\mathbf{M}_0\mathbf{v} = \mathbf{v} - \bar{v}\mathbf{1}$, so

$$\mathbf{v}'\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0\mathbf{v})'(\mathbf{M}_0\mathbf{v}) = \|\mathbf{v} - \bar{v}\mathbf{1}\|^2 = \sum_{i=1}^{n}(v_i - \bar{v})^2,$$

using $\mathbf{M}_0'\mathbf{M}_0 = \mathbf{M}_0^2 = \mathbf{M}_0$ (symmetric idempotent). Applying this with $\mathbf{v} = \mathbf{Y}$ gives $\mathrm{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y}$; with $\mathbf{v} = \hat{\mathbf{Y}}$ gives $\mathrm{SSR} = \hat{\mathbf{Y}}'\mathbf{M}_0\hat{\mathbf{Y}}$, provided $\overline{\hat{Y}} = \bar{Y}$—which follows from $\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.4(ii)) since $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'(\mathbf{Y} - \mathbf{e}) = \mathbf{1}'\mathbf{Y}$.

Step 2: Decompose $\mathbf{M}_0\mathbf{Y}$ into orthogonal components. Write

$$\mathbf{M}_0\mathbf{Y} = \mathbf{M}_0(\hat{\mathbf{Y}} + \mathbf{e}) = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{M}_0\mathbf{e} = \mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}, \tag{21}$$

where the last equality uses $\mathbf{M}_0\mathbf{e} = \mathbf{e} - \bar{e}\mathbf{1} = \mathbf{e}$ because $\bar{e} = n^{-1}\mathbf{1}'\mathbf{e} = 0$ (Theorem 2.4(ii)).

Step 3: Expand SST**.** Using Step 2 and $\mathbf{M}_0^2 = \mathbf{M}_0$:

$$\begin{aligned}\mathrm{SST} = \mathbf{Y}'\mathbf{M}_0\mathbf{Y} = (\mathbf{M}_0\mathbf{Y})'(\mathbf{M}_0\mathbf{Y}) &= (\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e})'(\mathbf{M}_0\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \mathrm{SSR} + 2\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} + \mathrm{SSE}.\end{aligned}$$

Step 4: The cross term vanishes. Using $\mathbf{e}'\hat{\mathbf{Y}} = 0$ and $\mathbf{e}'\mathbf{1} = 0$ (Theorem 2.4(ii), (iii)):

$$\mathbf{e}'\mathbf{M}_0\hat{\mathbf{Y}} = \mathbf{e}'\hat{\mathbf{Y}} - \bar{Y}\,\mathbf{e}'\mathbf{1} = 0.$$

Hence $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$. □

Remark (Geometric view)**.** Equation (21) is an orthogonal decomposition of $\mathbf{M}_0\mathbf{Y}$ into two pieces: the centered fitted values $\mathbf{M}_0\hat{\mathbf{Y}}$, which lie in the subspace $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ (the non-intercept directions of the column space), and the residuals $\mathbf{e}$, which lie in $\mathcal{C}(\mathbf{X})^\perp$. Pythagoras applied to these two orthogonal pieces is precisely $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$.

6.2 Coefficient of Determination

Definition 6.2 (R^2 and Adjusted R^2)**.** The **coefficient of determination** is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}. \tag{22}$$

It measures the proportion of total variation in $\mathbf{Y}$ explained by the regression. The **adjusted** R^2 is

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{n-1}{n-k-1}(1-R^2) = 1 - \frac{\text{MSE}}{s_Y^2}, \tag{23}$$

where $s_Y^2 = \text{SST}/(n-1)$. The adjusted R^2 can be negative.

Proposition 6.1 (Bounds on R^2)**.** In any model with an intercept, $0 \leqslant R^2 \leqslant 1$.

Proof. $R^2 = \text{SSR}/\text{SST} \geqslant 0$ because $\text{SSR} = \sum(\hat{Y}_i - \bar{Y})^2 \geqslant 0$ and $\text{SST} = \sum(Y_i - \bar{Y})^2 > 0$ (assuming $\mathbf{Y}$ is not constant). For the upper bound, Theorem 6.1 gives $\text{SST} = \text{SSR} + \text{SSE}$ with $\text{SSE} \geqslant 0$, so $\text{SSR} \leqslant \text{SST}$, i.e., $R^2 \leqslant 1$. □

Theorem 6.2 (Monotonicity of R^2)**.** Adding a regressor to the model cannot decrease R^2. That is, if R_k^2 is the coefficient of determination with k regressors and R_{k+1}^2 is that with an additional regressor, then $R_{k+1}^2 \geqslant R_k^2$.

Proof. Let $\mathcal{C}_k \subseteq \mathcal{C}_{k+1}$ denote the column spaces of the smaller and larger design matrices (the inclusion holds because every column of $\mathbf{X}_k$ is also a column of $\mathbf{X}_{k+1}$). Write $\hat{\mathbf{Y}}_k, \hat{\mathbf{Y}}_{k+1}$ for the corresponding OLS fits.

By Theorem 2.2, $\hat{\mathbf{Y}}_{k+1}$ is the point in $\mathcal{C}_{k+1}$ closest to $\mathbf{Y}$, so $\|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\| \leqslant \|\mathbf{Y} - \mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{C}_{k+1}$. In particular, $\hat{\mathbf{Y}}_k \in \mathcal{C}_k \subseteq \mathcal{C}_{k+1}$, so

$$\text{SSE}_{k+1} = \|\mathbf{Y} - \hat{\mathbf{Y}}_{k+1}\|^2 \;\leqslant\; \|\mathbf{Y} - \hat{\mathbf{Y}}_k\|^2 = \text{SSE}_k.$$

Since SST does not depend on the regressors,

$$R_{k+1}^2 = 1 - \frac{\text{SSE}_{k+1}}{\text{SST}} \;\geqslant\; 1 - \frac{\text{SSE}_k}{\text{SST}} = R_k^2.$$ □

Lemma 6.1 (Reduction in SSE from adding one regressor)**.** Write the enlarged design as $\mathbf{X}_{k+1} = [\mathbf{X}_k \;\; \mathbf{x}]$, where $\mathbf{x}$ is the new regressor column, and assume $\mathbf{x}'\mathbf{M}_k\mathbf{x} > 0$ (where $\mathbf{M}_k$ is the residual maker for $\mathbf{X}_k$), which holds whenever $\mathbf{x}$ is not in $\mathcal{C}(\mathbf{X}_k)$. Then the OLS coefficient on $\mathbf{x}$ in the enlarged model is

$$\hat{\beta}_{k+1} = \frac{\mathbf{x}'\mathbf{M}_k\mathbf{Y}}{\mathbf{x}'\mathbf{M}_k\mathbf{x}}, \tag{24}$$

and the reduction in residual sum of squares is

$$\text{SSE}_k - \text{SSE}_{k+1} = \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x} = \frac{(\mathbf{x}'\mathbf{M}_k\mathbf{Y})^2}{\mathbf{x}'\mathbf{M}_k\mathbf{x}} \;\geqslant\; 0. \tag{25}$$

This identity uses only normal equations (no forward dependence on Section 8).

Proof. The partitioned normal equations $\mathbf{X}_{k+1}'\mathbf{X}_{k+1}\hat{\boldsymbol{\beta}}_{k+1} = \mathbf{X}_{k+1}'\mathbf{Y}$ read

$$\mathbf{X}_k'\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{X}_k'\mathbf{x}\hat{\beta}_{k+1} = \mathbf{X}_k'\mathbf{Y}, \tag{a}$$

$$\mathbf{x}'\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}, \tag{b}$$

where $\hat{\boldsymbol{\beta}}_k^*$ denotes the first-k block of the enlarged estimator (generally different from the short-regression estimator $\hat{\boldsymbol{\beta}}_k$). From (a), $\hat{\boldsymbol{\beta}}_k^* = (\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'(\mathbf{Y} - \mathbf{x}\hat{\beta}_{k+1})$. Substituting into (b):

$$\mathbf{x}'\mathbf{X}_k(\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'(\mathbf{Y} - \mathbf{x}\hat{\beta}_{k+1}) + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y}.$$

Writing $\mathbf{H}_k = \mathbf{X}_k(\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k' = \mathbf{I} - \mathbf{M}_k$ and rearranging:

$$\mathbf{x}'\mathbf{H}_k\mathbf{Y} - \mathbf{x}'\mathbf{H}_k\mathbf{x}\,\hat{\beta}_{k+1} + \mathbf{x}'\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'\mathbf{Y} \iff \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{x}\,\hat{\beta}_{k+1} = \mathbf{x}'(\mathbf{I} - \mathbf{H}_k)\mathbf{Y},$$

which gives (24).

For (25), relate $\hat{\boldsymbol{\beta}}_k^*$ to the short-regression estimator $\hat{\boldsymbol{\beta}}_k = (\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{Y}$: from (a),

$$\hat{\boldsymbol{\beta}}_k^* = \hat{\boldsymbol{\beta}}_k - (\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{x}\,\hat{\beta}_{k+1}.$$

Multiplying by $\mathbf{X}_k$ gives $\mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* = \hat{\mathbf{Y}}_k - \mathbf{H}_k\mathbf{x}\,\hat{\beta}_{k+1}$, so

$$\hat{\mathbf{Y}}_{k+1} = \mathbf{X}_k\hat{\boldsymbol{\beta}}_k^* + \mathbf{x}\hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + (\mathbf{I} - \mathbf{H}_k)\mathbf{x}\,\hat{\beta}_{k+1} = \hat{\mathbf{Y}}_k + \mathbf{M}_k\mathbf{x}\,\hat{\beta}_{k+1},$$

and the residuals satisfy $\mathbf{e}_{k+1} = \mathbf{e}_k - \mathbf{M}_k\mathbf{x}\,\hat{\beta}_{k+1}$. Taking squared norms, using $\mathbf{e}_k = \mathbf{M}_k\mathbf{Y}$ and $\mathbf{M}_k^2 = \mathbf{M}_k$:

$$\mathrm{SSE}_{k+1} = \mathrm{SSE}_k - 2\hat{\beta}_{k+1} \cdot \mathbf{Y}'\mathbf{M}_k\mathbf{x} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}.$$

Substituting $\hat{\beta}_{k+1} = (\mathbf{x}'\mathbf{M}_k\mathbf{Y})/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$ makes the middle term $-2\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}$, so

$$\mathrm{SSE}_{k+1} = \mathrm{SSE}_k - \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x},$$

which is (25). The non-negativity reflects Theorem 6.2. □

Theorem 6.3 ($\bar{R}^2$ Increases iff $|t| > 1$)**.** Adding a single regressor increases $\bar{R}^2$ if and only if its t-statistic exceeds 1 in absolute value.

Proof. Let the model with k regressors have $\mathrm{MSE}_k = \mathrm{SSE}_k/(n-k-1)$, and the model with an additional regressor have $\mathrm{MSE}_{k+1} = \mathrm{SSE}_{k+1}/(n-k-2)$. Then $\bar{R}_{k+1}^2 > \bar{R}_k^2$ if and only if $\mathrm{MSE}_{k+1} < \mathrm{MSE}_k$, i.e.,

$$\frac{\mathrm{SSE}_{k+1}}{n-k-2} < \frac{\mathrm{SSE}_k}{n-k-1}. \tag{26}$$

Step 1: Apply Lemma 6.1. By (25), $\mathrm{SSE}_k = \mathrm{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}$. Substitute this into (26):

$$\begin{aligned}
\frac{\mathrm{SSE}_{k+1}}{n-k-2} &< \frac{\mathrm{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{n-k-1} \\
(n-k-1)\mathrm{SSE}_{k+1} &< (n-k-2)\big[\mathrm{SSE}_{k+1} + \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}\big] \\
\mathrm{SSE}_{k+1} &< (n-k-2) \cdot \hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}.
\end{aligned}$$

Dividing by $\mathrm{SSE}_{k+1} = (n-k-2)\mathrm{MSE}_{k+1} > 0$:

$$1 < \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\mathrm{MSE}_{k+1}}. \tag{27}$$

Step 2: Identify the right-hand side as t^2. The t-statistic for $\hat{\beta}_{k+1}$ in the enlarged model is $t_{k+1} = \hat{\beta}_{k+1}/\mathrm{SE}(\hat{\beta}_{k+1})$ with $\mathrm{SE}(\hat{\beta}_{k+1})^2 = \mathrm{MSE}_{k+1} \cdot [(\mathbf{X}_{k+1}'\mathbf{X}_{k+1})^{-1}]_{k+1,k+1}$ (defined formally in Theorem 7.1). By the block-inverse (Schur complement) formula applied to $\mathbf{X}_{k+1}'\mathbf{X}_{k+1}$:

$$[(\mathbf{X}_{k+1}'\mathbf{X}_{k+1})^{-1}]_{k+1,k+1} = (\mathbf{x}'\mathbf{x} - \mathbf{x}'\mathbf{X}_k(\mathbf{X}_k'\mathbf{X}_k)^{-1}\mathbf{X}_k'\mathbf{x})^{-1} = (\mathbf{x}'\mathbf{M}_k\mathbf{x})^{-1}.$$

Hence $\mathrm{SE}(\hat{\beta}_{k+1})^2 = \mathrm{MSE}_{k+1}/(\mathbf{x}'\mathbf{M}_k\mathbf{x})$, so

$$t_{k+1}^2 = \frac{\hat{\beta}_{k+1}^2}{\mathrm{SE}(\hat{\beta}_{k+1})^2} = \frac{\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}}{\mathrm{MSE}_{k+1}}.$$

Substituting into (27) gives $t_{k+1}^2 > 1$, i.e., $|t_{k+1}| > 1$. □

Remark (Logical status of the forward references)**.** The proof invokes two results proved later: the Schur complement identity for $(\mathbf{X}_{k+1}'\mathbf{X}_{k+1})^{-1}$, and the form of $\mathrm{SE}(\hat{\beta}_{k+1})$ from Theorem 7.1. Both are used only to *interpret* the ratio $\hat{\beta}_{k+1}^2 \cdot \mathbf{x}'\mathbf{M}_k\mathbf{x}/\mathrm{MSE}_{k+1}$ as t_{k+1}^2; equation (27) itself is a self-contained algebraic consequence of Lemma 6.1. A reader seeking strict linearity can defer Theorem 6.3 until after Section 7.

7 Hypothesis Testing and Confidence Regions

7.1 Individual Coefficient t Test

Theorem 7.1 (Individual t Test)**.** Under (A1)–(A5), for testing $H_0\colon \beta_j = \beta_j^0$:

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-k-1} \quad \text{under } H_0, \tag{28}$$

where $\mathrm{SE}(\hat{\beta}_j) = s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ and $s = \sqrt{\mathrm{MSE}}$.

Proof. Step 1: Marginal distribution of $\hat{\beta}_j$. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. The j-th coordinate of a multivariate normal vector is itself normal, with mean β_j and variance equal to the (j,j)-entry of the covariance matrix (this is Theorem 4.1 applied with $\mathbf{A} = \mathbf{e}_j'$, the j-th standard basis row vector):

$$\hat{\beta}_j = \mathbf{e}_j'\hat{\boldsymbol{\beta}} \sim N\big(\beta_j,\ \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}\big).$$

Standardizing:

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0,1).$$

Step 2: Chi-square quantity independent of Z. By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and is independent of $\hat{\boldsymbol{\beta}}$. Since Z is a (deterministic) function of $\hat{\beta}_j$, which is a coordinate of $\hat{\boldsymbol{\beta}}$, and a function of an independent random variable remains independent, Z is independent of V.

Step 3. By definition, if $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Applying this:

$$\begin{aligned}\frac{Z}{\sqrt{V/(n-k-1)}} &= \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \cdot \frac{1}{\sqrt{\mathrm{SSE}/[\sigma^2(n-k-1)]}} \\ &= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\mathrm{MSE}} \cdot \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} = \frac{\hat{\beta}_j - \beta_j}{s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim t_{n-k-1}.\end{aligned}$$

Under $H_0\colon \beta_j = \beta_j^0$, replace β_j by β_j^0. □

7.2 Overall F Test

Lemma 7.1 (Distribution of SSR/σ^2 Under H_0)**.** Define $\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}'$. Under $H_0\colon \beta_1 = \cdots = \beta_k = 0$ (with (A1)–(A5)):

$$\frac{\mathrm{SSR}}{\sigma^2} \sim \chi^2_k, \qquad \text{and SSR is independent of SSE.}$$

Proof. Step 1: Express SSR/σ^2 as a quadratic form in standard normals. Two matrix observations are useful:

- $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ by definition of the hat matrix.
- $\bar{Y}\mathbf{1} = \mathbf{H}_0\mathbf{Y}$, because $\mathbf{H}_0\mathbf{Y} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{Y}) = n^{-1}\mathbf{1} \cdot n\bar{Y} = \bar{Y}\mathbf{1}$.

Therefore

$$\hat{\mathbf{Y}} - \bar{Y}\mathbf{1} = \mathbf{H}\mathbf{Y} - \mathbf{H}_0\mathbf{Y} = (\mathbf{H} - \mathbf{H}_0)\mathbf{Y},$$

and $\mathrm{SSR} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})'(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_0)'(\mathbf{H} - \mathbf{H}_0)\mathbf{Y}$.

Under H_0, $\boldsymbol{\beta} = (\beta_0, 0, \ldots, 0)'$, so $\mathbf{X}\boldsymbol{\beta} = \beta_0\mathbf{1}$ and $\mathbf{Y} = \beta_0\mathbf{1} + \boldsymbol{\varepsilon}$. We claim $(\mathbf{H} - \mathbf{H}_0)\mathbf{Y} = (\mathbf{H} - \mathbf{H}_0)\boldsymbol{\varepsilon}$: indeed, since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ we have $\mathbf{H}\mathbf{1} = \mathbf{1}$ (by Theorem 2.3(ii) applied to the first column of $\mathbf{X}$), and $\mathbf{H}_0\mathbf{1} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1}$, so

$$(\mathbf{H} - \mathbf{H}_0)(\beta_0\mathbf{1}) = \beta_0(\mathbf{H}\mathbf{1} - \mathbf{H}_0\mathbf{1}) = \beta_0(\mathbf{1} - \mathbf{1}) = \mathbf{0},$$

and hence $(\mathbf{H} - \mathbf{H}_0)\mathbf{Y} = (\mathbf{H} - \mathbf{H}_0)(\beta_0\mathbf{1} + \boldsymbol{\varepsilon}) = (\mathbf{H} - \mathbf{H}_0)\boldsymbol{\varepsilon}$.

Let $\mathbf{z} = \boldsymbol{\varepsilon}/\sigma$. Under (A5), $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$. Then

$$\frac{\mathrm{SSR}}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'(\mathbf{H} - \mathbf{H}_0)'(\mathbf{H} - \mathbf{H}_0)\boldsymbol{\varepsilon}}{\sigma^2} = \mathbf{z}'(\mathbf{H} - \mathbf{H}_0)'(\mathbf{H} - \mathbf{H}_0)\mathbf{z}.$$

This reduces to $\mathbf{z}'(\mathbf{H} - \mathbf{H}_0)\mathbf{z}$ once we establish in Step 2 that $\mathbf{H} - \mathbf{H}_0$ is symmetric idempotent, which implies $(\mathbf{H} - \mathbf{H}_0)'(\mathbf{H} - \mathbf{H}_0) = (\mathbf{H} - \mathbf{H}_0)^2 = \mathbf{H} - \mathbf{H}_0$.

Step 2: $\mathbf{H} - \mathbf{H}_0$ is symmetric idempotent with rank k. Symmetry is immediate. For idempotency, we first show

$$\mathbf{H}\mathbf{H}_0 = \mathbf{H}_0 = \mathbf{H}_0\mathbf{H}. \tag{29}$$

Using $\mathbf{H}\mathbf{1} = \mathbf{1}$: $\mathbf{H}\mathbf{H}_0 = n^{-1}\mathbf{H}\mathbf{1}\mathbf{1}' = n^{-1}\mathbf{1}\mathbf{1}' = \mathbf{H}_0$; by symmetry, $\mathbf{H}_0\mathbf{H} = (\mathbf{H}\mathbf{H}_0)' = \mathbf{H}_0$. Also $\mathbf{H}_0^2 = n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{H}_0$. Then

$$(\mathbf{H} - \mathbf{H}_0)^2 = \mathbf{H}^2 - \mathbf{H}\mathbf{H}_0 - \mathbf{H}_0\mathbf{H} + \mathbf{H}_0^2 = \mathbf{H} - \mathbf{H}_0 - \mathbf{H}_0 + \mathbf{H}_0 = \mathbf{H} - \mathbf{H}_0.$$

For the rank, Lemma 2.2 gives

$$\mathrm{rank}(\mathbf{H} - \mathbf{H}_0) = \mathrm{tr}(\mathbf{H}) - \mathrm{tr}(\mathbf{H}_0) = (k + 1) - 1 = k,$$

with $\mathrm{tr}(\mathbf{H}_0) = n^{-1}\,\mathrm{tr}(\mathbf{1}\mathbf{1}') = 1$.

The Step 1 expression simplifies: $(\mathbf{H} - \mathbf{H}_0)'(\mathbf{H} - \mathbf{H}_0) = \mathbf{H} - \mathbf{H}_0$, so $\mathrm{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{H}_0)\mathbf{z} \sim \chi^2_k$ by Theorem 4.3.

Step 3: Independence of SSR and SSE. Apply Cochran's theorem to $\mathbf{A}_1 = \mathbf{H} - \mathbf{H}_0$ and $\mathbf{A}_2 = \mathbf{M}$. The required $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$ follows from $\mathbf{H}\mathbf{M} = \mathbf{0}$ (Theorem 2.3(iii)) and $\mathbf{H}_0\mathbf{M} = \mathbf{H}_0 - \mathbf{H}_0\mathbf{H} = \mathbf{0}$ by (29):

$$(\mathbf{H} - \mathbf{H}_0)\mathbf{M} = \mathbf{0}.$$

Hence $\mathrm{SSR}/\sigma^2 = \mathbf{z}'(\mathbf{H} - \mathbf{H}_0)\mathbf{z}$ and $\mathrm{SSE}/\sigma^2 = \mathbf{z}'\mathbf{M}\mathbf{z}$ are independent, and so are SSR and SSE. □

Theorem 7.2 (Overall F Test)**.** Under (A1)–(A5), for testing $H_0\colon \beta_1 = \beta_2 = \cdots = \beta_k = 0$:

$$\boxed{F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,\,n-k-1} \quad \text{under } H_0.} \tag{30}$$

Proof. By Lemma 7.1 and Theorem 5.2, under H_0: $\mathrm{SSR}/\sigma^2 \sim \chi^2_k$, $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$, and they are independent. By the definition of the F distribution:

$$F = \frac{(\mathrm{SSR}/\sigma^2)/k}{(\mathrm{SSE}/\sigma^2)/(n-k-1)} = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{\mathrm{MSR}}{\mathrm{MSE}} \sim F_{k,n-k-1}.$$

The equivalence $F = \frac{R^2/k}{(1-R^2)/(n-k-1)}$ follows from $R^2 = \mathrm{SSR}/\mathrm{SST}$ and $1 - R^2 = \mathrm{SSE}/\mathrm{SST}$. □

7.3 General Linear Hypothesis

Theorem 7.3 (General F Test)**.** Under (A1)–(A5), consider $H_0\colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ where $\mathbf{R}$ is a $q \times (k+1)$ matrix of rank q and $\mathbf{r} \in \mathbb{R}^q$. The test statistic

$$\boxed{F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q,\,n-k-1} \quad \text{under } H_0.} \tag{31}$$

Proof. Step 1. By Theorem 5.1, $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. By Theorem 4.1:

$$\mathbf{R}\hat{\boldsymbol{\beta}} \sim N\big(\mathbf{R}\boldsymbol{\beta},\ \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big).$$

Under H_0, $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r} \sim N(\mathbf{0}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}')$.

Step 2: A quadratic form in $\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}$ is chi-square. Let $\boldsymbol{\Sigma}_R = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. $\boldsymbol{\Sigma}_R$ is positive definite: for nonzero $\mathbf{v} \in \mathbb{R}^q$,

$$\mathbf{v}'\boldsymbol{\Sigma}_R\mathbf{v} = (\mathbf{R}'\mathbf{v})'(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{R}'\mathbf{v}) > 0,$$

since $\operatorname{rank}(\mathbf{R}) = q$ makes $\mathbf{R}'\mathbf{v} \neq \mathbf{0}$, and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite. Let $\boldsymbol{\Sigma}_R^{1/2}$ be its positive definite symmetric square root.

Set $\mathbf{w} = \sigma^{-1}\boldsymbol{\Sigma}_R^{-1/2}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$. Theorem 4.1 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_q)$, so $\mathbf{w}'\mathbf{w} \sim \chi^2_q$ by Theorem 4.3:

$$W \;:=\; \mathbf{w}'\mathbf{w} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{\sigma^2} \sim \chi^2_q.$$

Step 3. By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$, independent of $\hat{\boldsymbol{\beta}}$; since W is a function of $\hat{\boldsymbol{\beta}}$, W is independent of V. Then:

$$F = \frac{W/q}{V/(n-k-1)} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/(q\sigma^2)}{\mathrm{SSE}/[\sigma^2(n-k-1)]} = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\boldsymbol{\Sigma}_R^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q \cdot s^2} \sim F_{q,n-k-1}. \qquad \square$$

Remark (Special cases)**.** The individual t test (Theorem 7.1) is the case $q = 1$, $\mathbf{R} = \mathbf{e}_j'$ (the j-th standard basis vector), $\mathbf{r} = \beta_j^0$, where $t^2 = F \sim F_{1,n-k-1}$. The overall F test (Theorem 7.2) is the case $\mathbf{R} = [\mathbf{0}_{k\times 1} \mid \mathbf{I}_k]$, $\mathbf{r} = \mathbf{0}_k$.

Corollary 7.1 (Nested F Test via Residual Sum of Squares)**.** Let SSE_U and SSE_R be the residual sums of squares from the full and reduced models under $H_0\colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q constraints). Then the F statistic in Theorem 7.3 is equivalently expressed as

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/q}{\mathrm{SSE}_U/(n-k-1)}. \tag{32}$$

Proof. By Theorem 10.1, the constrained estimator satisfies

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \tag{33}$$

Step 1: Decompose SSE_R. Since $\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R) = \mathbf{e} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)$,

$$\begin{aligned}\mathrm{SSE}_R = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_R\|^2 &= \|\mathbf{e}\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 + 2\mathbf{e}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\\ &= \mathrm{SSE}_U + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2,\end{aligned}$$

since the cross term vanishes: $\mathbf{e}'\mathbf{X} = \mathbf{0}'$ by Theorem 2.4(i).

Step 2: Simplify the squared norm. Let $\mathbf{A} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})$ denote the trailing factor in (33), so that $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}$. Then

$$\begin{aligned}
\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 &= (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R) \\
&= \big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}\big]'\mathbf{X}'\mathbf{X}\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}\big] \\
&= \mathbf{A}'\mathbf{R}\underbrace{(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}}_{=(\mathbf{X}'\mathbf{X})^{-1}}\mathbf{R}'\mathbf{A} \\
&= \mathbf{A}'\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\mathbf{A}.
\end{aligned}$$

Expanding $\mathbf{A}$:

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}\underbrace{\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}}_{=\mathbf{I}_q}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

The bracketed product collapses to $\mathbf{I}_q$, leaving

$$\mathrm{SSE}_R - \mathrm{SSE}_U = \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_R)\|^2 = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}). \tag{34}$$

Step 3: Compare with Theorem 7.3. The numerator in Theorem 7.3's F statistic equals the right-hand side of (34) divided by q. Since the denominator $s^2 = \mathrm{SSE}_U/(n-k-1)$ is the same, substituting yields

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/q}{\mathrm{SSE}_U/(n-k-1)}. \qquad \square$$

Remark (Logical status of the forward reference)**.** The proof relies on the constrained estimator formula (33) from Theorem 10.1 in Section 10. The algebraic content of (34)—that $\mathrm{SSE}_R - \mathrm{SSE}_U$ admits the quadratic form on the right—can alternatively be derived without constrained OLS via the orthogonal decomposition $\mathbf{Y} = \mathbf{H}_R\mathbf{Y} + (\mathbf{H} - \mathbf{H}_R)\mathbf{Y} + \mathbf{M}\mathbf{Y}$, where $\mathbf{H}_R$ is the projection onto the restricted column space and $\mathbf{H} - \mathbf{H}_R$ is idempotent of rank q (so $\mathrm{SSE}_R - \mathrm{SSE}_U = \mathbf{Y}'(\mathbf{H} - \mathbf{H}_R)\mathbf{Y}$ by Pythagoras). Constrained OLS is needed only to give the explicit form $\mathbf{H}_R = \mathbf{H} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. A reader seeking strict linearity can defer this corollary until after Section 10.

Corollary 7.2 ($t^2 = F$ for a Single Linear Constraint)**.** For testing $H_0\colon \mathbf{c}'\boldsymbol{\beta} = r_0$ (a single constraint, $q = 1$), the F statistic from Theorem 7.3 equals the square of the t statistic:

$$t = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}, \qquad F = t^2 \sim F_{1,n-k-1}.$$

Proof. With $q = 1$, $\mathbf{R} = \mathbf{c}'$, $\mathbf{r} = r_0$, the general F statistic becomes:

$$F = \frac{(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)[\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}]^{-1}(\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0)}{1 \cdot s^2} = \left(\frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - r_0}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}}\right)^2 = t^2. \qquad \square$$

7.4 Confidence Intervals and Regions

Theorem 7.4 (Confidence Interval for $\mathbf{c}'\boldsymbol{\beta}$)**.** Under (A1)–(A5), a $(1-\alpha)\times 100\%$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-k-1} \cdot s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}, \tag{35}$$

where $t_{\alpha/2,n-k-1}$ is the upper $\alpha/2$ quantile of t_{n-k-1}.

Proof. By Theorem 4.1 applied to $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$ with $\mathbf{A} = \mathbf{c}'$ (a $1 \times (k+1)$ row vector), $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta},\ \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. Standardizing:

$$Z = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim N(0,1).$$

By Theorem 5.2, $V = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and is independent of $\hat{\boldsymbol{\beta}}$; since Z is a function of $\hat{\boldsymbol{\beta}}$ alone, Z is independent of V. By definition of the t distribution:

$$\frac{Z}{\sqrt{V/(n-k-1)}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{\sigma\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \cdot \frac{1}{\sqrt{\mathrm{SSE}/[\sigma^2(n-k-1)]}} = \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \sim t_{n-k-1},$$

where the last step uses $s^2 = \mathrm{SSE}/(n-k-1)$. The pivot gives

$$\mathsf{P}\left(-t_{\alpha/2} \leqslant \frac{\mathbf{c}'\hat{\boldsymbol{\beta}} - \mathbf{c}'\boldsymbol{\beta}}{s\sqrt{\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}} \leqslant t_{\alpha/2}\right) = 1 - \alpha,$$

and rearranging for $\mathbf{c}'\boldsymbol{\beta}$ yields the interval. □

Theorem 7.5 (Joint Confidence Ellipsoid)**.** Under (A1)–(A5), a $(1-\alpha) \times 100\%$ joint confidence region for $\boldsymbol{\beta}$ is

$$\left\{\boldsymbol{\beta} \in \mathbb{R}^{k+1} : (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leqslant (k+1)s^2 F_{\alpha,k+1,n-k-1}\right\}. \tag{36}$$

Proof. Step 1: A chi-square quadratic form in $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$. By Theorem 5.1, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. Let $(\mathbf{X}'\mathbf{X})^{1/2}$ be the positive definite symmetric square root and define $\mathbf{w} = \sigma^{-1}(\mathbf{X}'\mathbf{X})^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Theorem 4.1 gives $\mathbf{w} \sim N(\mathbf{0}, \mathbf{I}_{k+1})$, so $\mathbf{w}'\mathbf{w} \sim \chi^2_{k+1}$ by Theorem 4.3:

$$\mathbf{w}'\mathbf{w} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{\sigma^2} \sim \chi^2_{k+1}.$$

Step 2: Form the F ratio. This is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ (Theorem 5.2). By definition of F,

$$\frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/[(k+1)\sigma^2]}{\mathrm{SSE}/[\sigma^2(n-k-1)]} = \frac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'(\mathbf{X}'\mathbf{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}{(k+1)s^2} \sim F_{k+1,n-k-1}.$$

The confidence region collects the $\boldsymbol{\beta}$ for which this statistic is at most $F_{\alpha,k+1,n-k-1}$. □

8 Partitioned Regression

8.1 The Frisch–Waugh–Lovell Theorem

Theorem 8.1 (Frisch–Waugh–Lovell)**.** Partition $\mathbf{X} = [\mathbf{X}_1 \ \ \mathbf{X}_2]$ and $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2')'$, so the model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$ be the residual maker for $\mathbf{X}_2$. Then:

(i) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y}$.

(ii) The residual vector $\mathbf{e}$ from the full regression equals the residual vector from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$.

Proof. (i). The normal equations for the full model are:

$$\begin{pmatrix} \mathbf{X}_1'\mathbf{X}_1 & \mathbf{X}_1'\mathbf{X}_2 \\ \mathbf{X}_2'\mathbf{X}_1 & \mathbf{X}_2'\mathbf{X}_2 \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}_1'\mathbf{Y} \\ \mathbf{X}_2'\mathbf{Y} \end{pmatrix}.$$

The second block equation reads $\mathbf{X}_2'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_2'\mathbf{Y}$, which rearranges to

$$\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1).$$

Substituting this into the first block equation $\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \mathbf{X}_1'\mathbf{Y}$:

$$\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \underbrace{\mathbf{X}_1'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'}_{=\mathbf{X}_1'\mathbf{H}_2}(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1'\mathbf{Y}, \tag{37}$$

where we recognize the projection onto $\mathcal{C}(\mathbf{X}_2)$:

$$\mathbf{H}_2 := \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2' = \mathbf{I} - \mathbf{M}_2.$$

Expanding (37) using $\mathbf{X}_1'\mathbf{H}_2 = \mathbf{X}_1' - \mathbf{X}_1'\mathbf{M}_2$:

$$\mathbf{X}_1'\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_1'(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) - \mathbf{X}_1'\mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{X}_1'\mathbf{Y}.$$

The first two terms combine to $\mathbf{X}_1'\mathbf{Y}$, cancelling the right-hand side, leaving $\mathbf{X}_1'\mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1) = \mathbf{0}$, or

$$\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}_1'\mathbf{M}_2\mathbf{Y}.$$

Since $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1 = (\mathbf{M}_2\mathbf{X}_1)'(\mathbf{M}_2\mathbf{X}_1)$ is positive definite (columns of $\mathbf{M}_2\mathbf{X}_1$ are linearly independent by (A4)):

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y}.$$

(ii). Let $\tilde{\mathbf{Y}} = \mathbf{M}_2\mathbf{Y}$ and $\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1$. The residual from regressing $\tilde{\mathbf{Y}}$ on $\tilde{\mathbf{X}}_1$ is

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{Y} - \mathbf{M}_2\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}_2(\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1).$$

Substituting $\mathbf{Y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{e}$ from the full regression and using $\mathbf{M}_2\mathbf{X}_2 = \mathbf{0}$:

$$\tilde{\mathbf{e}} = \mathbf{M}_2\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \mathbf{M}_2\mathbf{e} = \mathbf{M}_2\mathbf{e}. \tag{38}$$

Finally, $\mathbf{M}_2\mathbf{e} = \mathbf{e}$: since $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (Theorem 2.4(i)) includes $\mathbf{X}_2'\mathbf{e} = \mathbf{0}$,

$$\mathbf{M}_2\mathbf{e} = \mathbf{e} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{e} = \mathbf{e}.$$

Hence $\tilde{\mathbf{e}} = \mathbf{e}$. □

Remark (The key identity $\mathbf{M}_2\mathbf{M} = \mathbf{M}$)**.** The computation above rests on the identity $\mathbf{M}_2\mathbf{e} = \mathbf{e}$, which can be written equivalently as $\mathbf{M}_2\mathbf{M} = \mathbf{M}$ (applying both sides to $\mathbf{Y}$). This holds because $\mathcal{C}(\mathbf{X}_2) \subseteq \mathcal{C}(\mathbf{X})$, so the orthogonal complement $\mathcal{C}(\mathbf{X})^\perp \subseteq \mathcal{C}(\mathbf{X}_2)^\perp$: anything already orthogonal to $\mathbf{X}$ is automatically orthogonal to $\mathbf{X}_2$, and $\mathbf{M}_2$ leaves it unchanged. This identity is the abstract reason why "residualizing with respect to a subset of regressors" preserves the full-model residual.

8.2 Omitted Variable Bias

Theorem 8.2 (Omitted Variable Bias)**.** Suppose the true model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$, but the researcher estimates the "short" regression $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\gamma} + \mathbf{u}$, obtaining $\tilde{\boldsymbol{\gamma}} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y}$. Then:

$$\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + \underbrace{(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2}_{\text{auxiliary regression coefficients}} \boldsymbol{\beta}_2. \tag{39}$$

The estimator $\tilde{\boldsymbol{\gamma}}$ is unbiased for $\boldsymbol{\beta}_1$ if and only if $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which holds when either $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$ ($\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal) or $\boldsymbol{\beta}_2 = \mathbf{0}$ (the omitted variables are irrelevant).

Proof. Substituting the true DGP $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ into the short-regression estimator:

$$\begin{aligned}\tilde{\gamma} &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{Y} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\boldsymbol{\varepsilon}.\end{aligned}$$

Taking expectations (using $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$\mathsf{E}[\tilde{\gamma}] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2. \qquad \square$$

Remark (Two-factor structure of the bias)**.** The bias term $(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\,\boldsymbol{\beta}_2$ is the product of two conceptually distinct quantities:

- $(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2$ is the matrix of OLS coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$—an *auxiliary regression* describing how the omitted variables relate to the included ones.
- $\boldsymbol{\beta}_2$ is the effect of the omitted variables on $\mathbf{Y}$ in the true model.

The bias vanishes if either factor is zero: if $\mathbf{X}_1$ and $\mathbf{X}_2$ are orthogonal ($\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$), omission is harmless; if the omitted variables have no effect ($\boldsymbol{\beta}_2 = \mathbf{0}$), there is nothing to omit. Both ingredients must be nonzero to generate bias.

Remark (Connection to FWL)**.** Theorem 8.1 shows that the correct (full-model) coefficient $\hat{\boldsymbol{\beta}}_1$ is obtained by regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$—that is, after partialling $\mathbf{X}_2$ out of both $\mathbf{Y}$ and $\mathbf{X}_1$. The short regression instead regresses $\mathbf{Y}$ directly on $\mathbf{X}_1$, effectively replacing $\mathbf{M}_2$ with $\mathbf{I}$. The resulting discrepancy is precisely the bias formula (39): OVB measures the cost of not partialling out the omitted variables.

9 Prediction

Let $\mathbf{x}_0 \in \mathbb{R}^{k+1}$ be a new observation's regressor vector. The predicted value is $\hat{Y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

Theorem 9.1 (Confidence Interval for the Conditional Mean)**.** Under (A1)–(A5), a $(1-\alpha)\times 100\%$ confidence interval for $\mathsf{E}[Y_0 \mid \mathbf{x}_0] = \mathbf{x}_0'\boldsymbol{\beta}$ is

$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-k-1} \cdot s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \tag{40}$$

Proof. This is the special case $\mathbf{c} = \mathbf{x}_0$ of Theorem 7.4. $\square$

Theorem 9.2 (Prediction Interval for a New Observation)**.** Under (A1)–(A5), a $(1-\alpha)\times 100\%$ prediction interval for a new observation $Y_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0$ is

$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-k-1} \cdot s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}. \tag{41}$$

Proof. Step 1: Set up. The prediction error is

$$\hat{Y}_0 - Y_0 = \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \varepsilon_0.$$

Here $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ depends on the training-sample errors $\boldsymbol{\varepsilon}$, whereas ε_0 is a new-observation error. Extending (A3) and (A5) to ε_0, $\operatorname{cov}(\boldsymbol{\varepsilon}, \varepsilon_0) = \mathbf{0}$ with joint normality, so $\varepsilon_0 \perp \boldsymbol{\varepsilon}$ (Theorem 4.2) and hence $\varepsilon_0 \perp \mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$.

Step 2: Compute the variance. Using Step 1's independence and Lemma 3.1 (with $\mathbf{A} = \mathbf{x}_0'$):

$$\begin{aligned}\operatorname{var}(\hat{Y}_0 - Y_0) &= \operatorname{var}\big(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\big) + \operatorname{var}(\varepsilon_0) \\ &= \mathbf{x}_0'\operatorname{var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0 + \sigma^2 \\ &= \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 + \sigma^2 \\ &= \sigma^2\big(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0\big).\end{aligned}$$

Step 3: Derive the t pivotal quantity. As a linear combination of the normals $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and ε_0, the prediction error is normal:

$$\hat{Y}_0 - Y_0 \sim N\big(0,\ \sigma^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)\big).$$

Independence from SSE holds because $\hat{\boldsymbol{\beta}} \perp \text{SSE}$ (Theorem 5.2) and ε_0 is independent of $\boldsymbol{\varepsilon}$ (hence of SSE, a function of $\boldsymbol{\varepsilon}$). Replacing σ with s gives a t_{n-k-1} pivot:

$$\frac{\hat{Y}_0 - Y_0}{s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim t_{n-k-1},$$

and inverting yields the prediction interval. □

Remark. The prediction interval is always wider than the confidence interval for the mean, because the prediction variance includes the additional σ^2 term from the irreducible noise ε_0.

10 Constrained Least Squares and Modeling Remarks

10.1 Remarks on Model Specification

Remark (No-intercept model)**.** When the model omits the intercept, $\mathbf{X}$ has no column of ones. The OLS formula $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ still applies, but $\sum e_i \neq 0$ in general and the decomposition SST = SSR + SSE may fail, so the centered R^2 can be negative. Use the uncentered $R^2 = 1 - \text{SSE}/\sum Y_i^2$ instead.

Remark (Dummy variable trap)**.** A categorical variable with g levels requires $g-1$ dummy regressors when the model includes an intercept. Including all g dummies creates the linear dependence $\mathbf{1} = D_1 + \cdots + D_g$, violating (A4).

10.2 Constrained Least Squares

Theorem 10.1 (Constrained OLS)**.** Consider $\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ (q linear constraints, $\text{rank}(\mathbf{R}) = q$). The constrained estimator is:

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\big]^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}), \tag{42}$$

and $\text{SSE}_R \geqslant \text{SSE}_U$ (where SSE_U is the unconstrained residual sum of squares).

Proof. The objective $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ is a convex quadratic function of $\boldsymbol{\beta}$ (its Hessian $2\mathbf{X}'\mathbf{X}$ is positive definite by (A4)), and the constraint $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ is affine. Hence the KKT conditions are necessary and sufficient for a global minimum.

Step 1: Lagrangian and first-order conditions. The Lagrangian is $\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\lambda}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + 2\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})$ (the factor 2 simplifies later algebra). By Lemma 2.1, differentiating $(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{Y}'\mathbf{Y} - 2\mathbf{Y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}'$ gives $-2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}$. The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\beta}'} = -2\mathbf{Y}'\mathbf{X} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{X} + 2\boldsymbol{\lambda}'\mathbf{R} = \mathbf{0}', \tag{i}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}'} = 2(\mathbf{R}\boldsymbol{\beta} - \mathbf{r})' = \mathbf{0}'. \tag{ii}$$

Transposing (i) gives $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_R + \mathbf{R}'\hat{\boldsymbol{\lambda}} = \mathbf{X}'\mathbf{Y}$, so

$$\hat{\boldsymbol{\beta}}_R = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}. \tag{43}$$

Step 2: Solve for $\hat{\boldsymbol{\lambda}}$. Left-multiply (43) by $\mathbf{R}$ and impose the constraint (ii), $\mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{r}$:

$$\mathbf{r} = \mathbf{R}\hat{\boldsymbol{\beta}}_R = \mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\hat{\boldsymbol{\lambda}}.$$

Since $\operatorname{rank}(\mathbf{R}) = q$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is positive definite, $\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$ is positive definite (hence invertible); see Step 2 of the proof of Theorem 7.3 for the detailed argument. Rearranging:

$$\hat{\boldsymbol{\lambda}} = [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

Substituting back into (43) gives (42).

Step 3: $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$**.** The constrained problem minimizes $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\{\boldsymbol{\beta} : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}\} \subseteq \mathbb{R}^{k+1}$. Since $\hat{\boldsymbol{\beta}}$ minimizes the same objective over the full space, the minimum over a subset cannot be smaller: $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. Equality holds iff $\hat{\boldsymbol{\beta}}$ already satisfies the constraint, i.e., $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{r}$. □

11 Exercises

Tier A: Reproduce the Derivation

Exercise 11.1. Establish the Magnus matrix differentiation formulas of Lemma 2.1: for constant $\mathbf{a} \in \mathbb{R}^p$ and constant $\mathbf{A} \in \mathbb{R}^{p\times p}$, prove $D(\mathbf{a}'\boldsymbol{\beta}) = \mathbf{a}'$ and $D(\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}) = \boldsymbol{\beta}'(\mathbf{A} + \mathbf{A}')$ via the differential rules and the first identification theorem $\mathrm{d}f(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'\mathrm{d}\boldsymbol{\beta} \Leftrightarrow Df(\boldsymbol{\beta}) = \mathbf{c}(\boldsymbol{\beta})'$.

Exercise 11.2. Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric and idempotent ($\mathbf{A}^2 = \mathbf{A} = \mathbf{A}'$). (a) Show every eigenvalue of $\mathbf{A}$ lies in $\{0, 1\}$. (b) Conclude $\operatorname{tr}(\mathbf{A}) = \operatorname{rank}(\mathbf{A})$. (c) Verify the formula on $\mathbf{H}$ and $\mathbf{M}$: $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Exercise 11.3. Derive the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ by minimizing $\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Verify the second-order condition.

Exercise 11.4. For $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M} = \mathbf{I} - \mathbf{H}$: (a) prove $\mathbf{H}$ and $\mathbf{M}$ are symmetric and idempotent; (b) prove $\mathbf{HX} = \mathbf{X}$, $\mathbf{MX} = \mathbf{0}$, and $\mathbf{HM} = \mathbf{0}$; (c) conclude $\operatorname{tr}(\mathbf{H}) = k + 1$ and $\operatorname{tr}(\mathbf{M}) = n - k - 1$.

Exercise 11.5. Prove that $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ under (A1)–(A2), and derive $\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under (A1)–(A4).

Exercise 11.6. State and prove the Gauss–Markov theorem.

Exercise 11.7. Prove that $\mathrm{MSE} = \mathrm{SSE}/(n-k-1)$ is an unbiased estimator of σ^2 under (A1)–(A4).

Exercise 11.8. Let $\mathbf{z} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. (a) For any constant matrix $\mathbf{A}$ and vector $\mathbf{b}$, prove $\mathbf{Az} + \mathbf{b} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b},\ \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$ via the moment generating function. (b) Specialise to $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}$ be symmetric idempotent of rank r; prove $\mathbf{z}'\mathbf{Az} \sim \chi^2_r$ via the spectral decomposition.

Exercise 11.9. (Cochran's theorem.) Let $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I}_n)$ and let $\mathbf{A}_1, \mathbf{A}_2$ be symmetric idempotent with $\mathbf{A}_1\mathbf{A}_2 = \mathbf{0}$. Prove that $\mathbf{z}'\mathbf{A}_1\mathbf{z}$ and $\mathbf{z}'\mathbf{A}_2\mathbf{z}$ are independent. *Hint:* use joint normality of $(\mathbf{A}_1\mathbf{z}, \mathbf{A}_2\mathbf{z})$ and the equivalence "uncorrelated $\Leftrightarrow$ independent" under joint normality.

Exercise 11.10. Under (A1)–(A5), prove $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-k-1}$ and that SSE is independent of $\hat{\boldsymbol{\beta}}$.

Exercise 11.11. Prove the variance decomposition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$, and derive the overall F statistic including the proof that $\mathrm{SSR}/\sigma^2 \sim \chi^2_k$ under $H_0\colon \beta_1 = \cdots = \beta_k = 0$.

Tier B: Apply the Theory

Exercise 11.12. Let $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. (a) Show $0 \leqslant h_{ii} \leqslant 1$. (b) Show $\sum_{i=1}^n h_{ii} = k + 1$. (c) Derive $\operatorname{var}(e_i) = \sigma^2(1 - h_{ii})$.

Exercise 11.13. Consider testing $H_0\colon \mathbf{c}'\boldsymbol{\beta} = r_0$ for a given vector $\mathbf{c}$. (a) Derive the distribution of $\mathbf{c}'\hat{\boldsymbol{\beta}}$. (b) Construct the t statistic and prove $t^2 \sim F_{1,n-k-1}$ under H_0.

Exercise 11.14. Consider two nested models: the full model with k regressors and a reduced model with $k_0 < k$ regressors. Let SSE_R and SSE_U denote the respective residual sums of squares. (a) Prove $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. (b) Show that $\frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/(k-k_0)}{\mathrm{SSE}_U/(n-k-1)} \sim F_{k-k_0,n-k-1}$ under H_0.

Exercise 11.15. (General linear hypothesis.) For $H_0\colon \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ a $q \times (k+1)$ matrix of rank q, derive the test statistic

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/q}{\mathrm{SSE}/(n-k-1)}$$

and prove $F \sim F_{q,n-k-1}$ under H_0.

Exercise 11.16. Suppose $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{\Omega}$ where $\boldsymbol{\Omega}$ is known, symmetric, and positive definite (but $\neq \mathbf{I}$). (a) Show that OLS is unbiased but derive its actual (incorrect) variance. (b) Define $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{Y}$ and prove it is BLUE.

Exercise 11.17. Let $\hat{Y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$ be the predicted value at a new point $\mathbf{x}_0$. (a) Derive $\mathrm{var}(\hat{Y}_0 - \mathsf{E}[Y_0])$. (b) Derive $\mathrm{var}(\hat{Y}_0 - Y_0)$. (c) Explain why (b) always exceeds (a).

Tier C: Extend and Connect

Exercise 11.18. (Frisch–Waugh–Lovell.) Partition $\mathbf{X} = [\mathbf{X}_1 \;\; \mathbf{X}_2]$. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$. Prove: (a) $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{Y}$. (b) The full regression residuals equal those from regressing $\mathbf{M}_2\mathbf{Y}$ on $\mathbf{M}_2\mathbf{X}_1$.

Exercise 11.19. (Omitted variable bias.) The true model is $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$, but only $\mathbf{Y} = \mathbf{X}_1\boldsymbol{\gamma} + \mathbf{u}$ is estimated. Show $\mathsf{E}[\tilde{\boldsymbol{\gamma}}] = \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2$. Under what conditions is $\tilde{\boldsymbol{\gamma}}$ unbiased?

Exercise 11.20. (Monotonicity of R^2 and the adjusted R^2 criterion.) (a) Prove that adding a regressor cannot decrease R^2. (b) Show that $\bar{R}^2$ increases if and only if the added variable's $|t|$-statistic exceeds 1.

Exercise 11.21. (Constrained least squares.) Consider $\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. (a) Derive the constrained estimator $\hat{\boldsymbol{\beta}}_R$ via Lagrange multipliers. (b) Show

$$\hat{\boldsymbol{\beta}}_R = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}).$$

(c) Prove $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$.

12 Solutions to Exercises

Solutions to Tier A

Solution to Exercise 11.1. See the proof of Lemma 2.1.

Solution to Exercise 11.2. See the proof of Lemma 2.2, and Theorem 2.3(iv) for part (c).

Solution to Exercise 11.3. See the proof of Theorem 2.1.

Solution to Exercise 11.4. See the proof of Theorem 2.3.

Solution to Exercise 11.5. See the proofs of Theorems 3.1 and 3.2.

Solution to Exercise 11.6. See the proof of Theorem 3.3.

Solution to Exercise 11.7. See the proof of Theorem 3.4.

Solution to Exercise 11.8. See the proofs of Theorems 4.1 and 4.3.

Solution to Exercise 11.9. See the proof of Theorem 4.4.

Solution to Exercise 11.10. See the proof of Theorem 5.2.

Solution to Exercise 11.11. See the proofs of Theorems 6.1, 7.2, and Lemma 7.1.

Solutions to Tier B

Solution to Exercise 11.12. See the proofs of Theorems 2.5 and 3.5.

Solution to Exercise 11.13. (a) By Theorem 4.1, $\mathbf{c}'\hat{\boldsymbol{\beta}} \sim N(\mathbf{c}'\boldsymbol{\beta},\ \sigma^2\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c})$. (b) See Corollary 7.2.

Solution to Exercise 11.14. (a) The reduced model minimizes SSE over a subspace $\{\boldsymbol{\beta} : \beta_{k_0+1} = \cdots = \beta_k = 0\}$. Since the unconstrained minimum is taken over the full space, $\mathrm{SSE}_R \geqslant \mathrm{SSE}_U$. (b) This is a special case of the general F test (Theorem 7.3) with $\mathbf{R} = [\mathbf{0} \mid \mathbf{I}_{k-k_0}]$ and $\mathbf{r} = \mathbf{0}$, or equivalently, by Cochran's theorem: define $\mathbf{M}_R$ and $\mathbf{M}_U$ as the residual makers for the reduced and full models. Then $\mathrm{SSE}_R - \mathrm{SSE}_U = \boldsymbol{\varepsilon}'(\mathbf{M}_R - \mathbf{M}_U)\boldsymbol{\varepsilon}/\sigma^2 \cdot \sigma^2$, where $\mathbf{M}_R - \mathbf{M}_U$ is symmetric idempotent of rank $k - k_0$ and $(\mathbf{M}_R - \mathbf{M}_U)\mathbf{M}_U = \mathbf{0}$. By Theorems 4.3 and 4.4, $(\mathrm{SSE}_R - \mathrm{SSE}_U)/\sigma^2 \sim \chi^2_{k-k_0}$, independent of $\mathrm{SSE}_U/\sigma^2 \sim \chi^2_{n-k-1}$. Hence $F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_U)/(k-k_0)}{\mathrm{SSE}_U/(n-k-1)} \sim F_{k-k_0,n-k-1}$.

Solution to Exercise 11.15. See the proof of Theorem 7.3.

Solution to Exercise 11.16. (a) Unbiasedness: $\mathsf{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathsf{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}$ (unchanged; unbiasedness does not use (A3)). Actual variance:

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\boldsymbol{\Omega})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \neq \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

(b) Let $\boldsymbol{\Omega} = \mathbf{P}\mathbf{P}'$ (Cholesky or symmetric square root). Transform: $\mathbf{P}^{-1}\mathbf{Y} = \mathbf{P}^{-1}\mathbf{X}\boldsymbol{\beta} + \mathbf{P}^{-1}\boldsymbol{\varepsilon}$, where $\mathrm{var}(\mathbf{P}^{-1}\boldsymbol{\varepsilon}) = \sigma^2\mathbf{P}^{-1}\boldsymbol{\Omega}(\mathbf{P}^{-1})' = \sigma^2\mathbf{I}$. The OLS estimator applied to the transformed model is

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = [(\mathbf{P}^{-1}\mathbf{X})'(\mathbf{P}^{-1}\mathbf{X})]^{-1}(\mathbf{P}^{-1}\mathbf{X})'(\mathbf{P}^{-1}\mathbf{Y}) = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{Y}.$$

Since the transformed model satisfies (A1)–(A4), Gauss–Markov (Theorem 3.3) guarantees $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}$ is BLUE in the transformed model. By a change-of-variables argument, it is also BLUE in the original model.

Solution to Exercise 11.17. See the proof of Theorem 9.2. (a) $\mathrm{var}(\hat{Y}_0 - \mathsf{E}[Y_0]) = \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$. (b) $\mathrm{var}(\hat{Y}_0 - Y_0) = \sigma^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)$. (c) the prediction variance exceeds the estimation variance by exactly σ^2, the irreducible noise variance of ε_0.

Solutions to Tier C

Solution to Exercise 11.18. See the proof of Theorem 8.1.

Solution to Exercise 11.19. See the proof of Theorem 8.2. Unbiasedness holds iff $\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$, which occurs when $\mathbf{X}_1 \perp \mathbf{X}_2$ or $\boldsymbol{\beta}_2 = \mathbf{0}$.

Solution to Exercise 11.20. See the proofs of Theorems 6.2 and 6.3.

Solution to Exercise 11.21. See the proof of Theorem 10.1.